

ECE 105: Introduction to Electrical Engineering

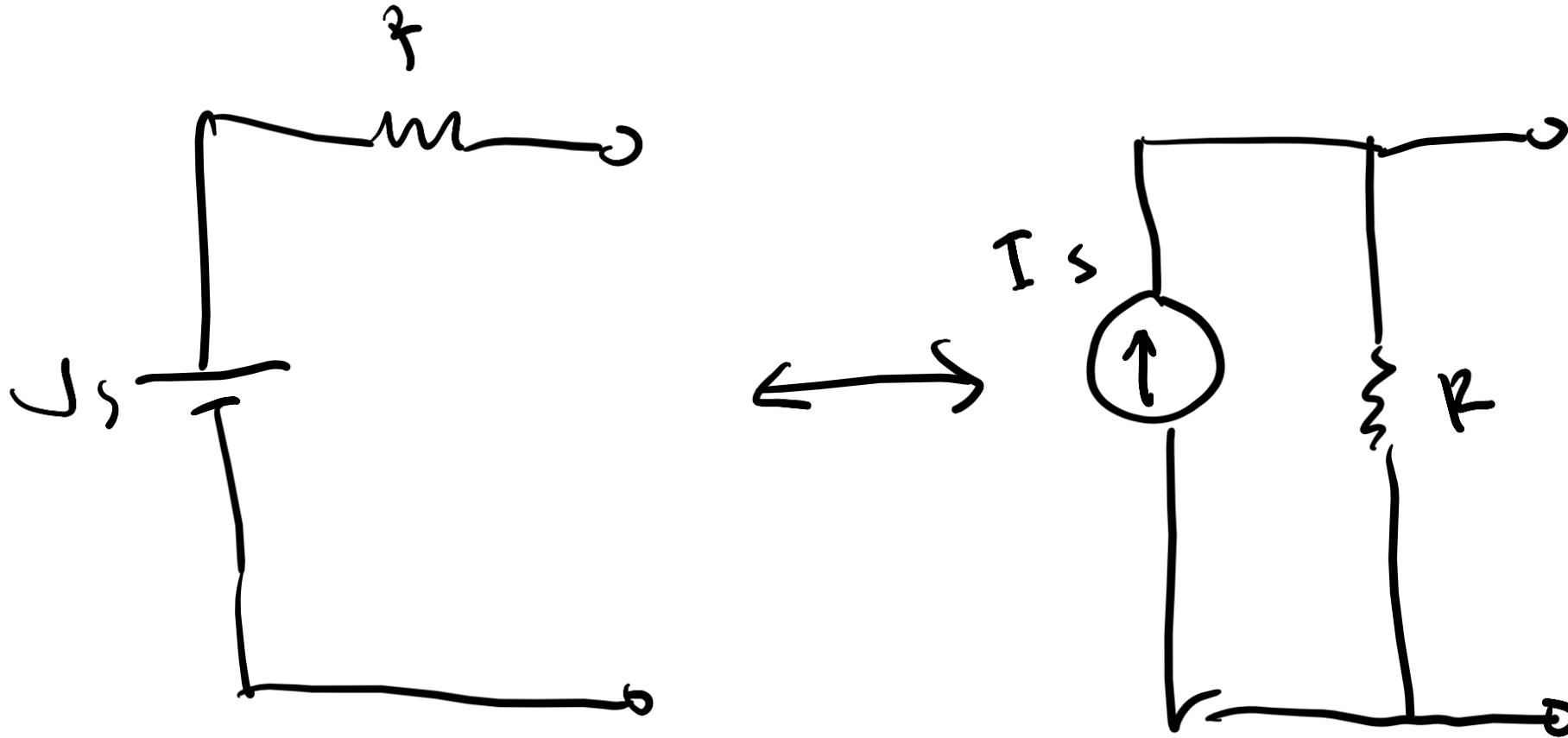
Lecture 7

Circuit Analysis - 3

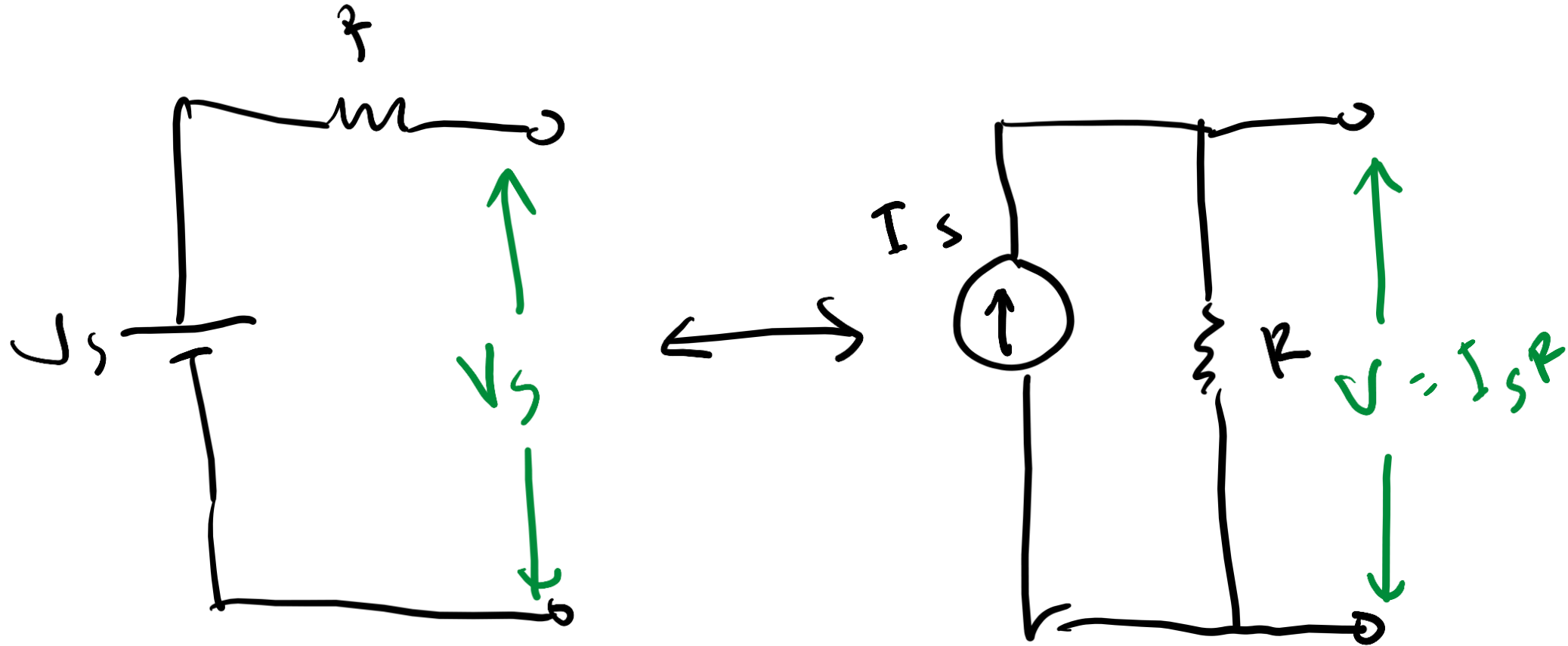
Yasser Khan

Rehan Kapadia

Source transformation – Voltage $\leftrightarrow$ Current



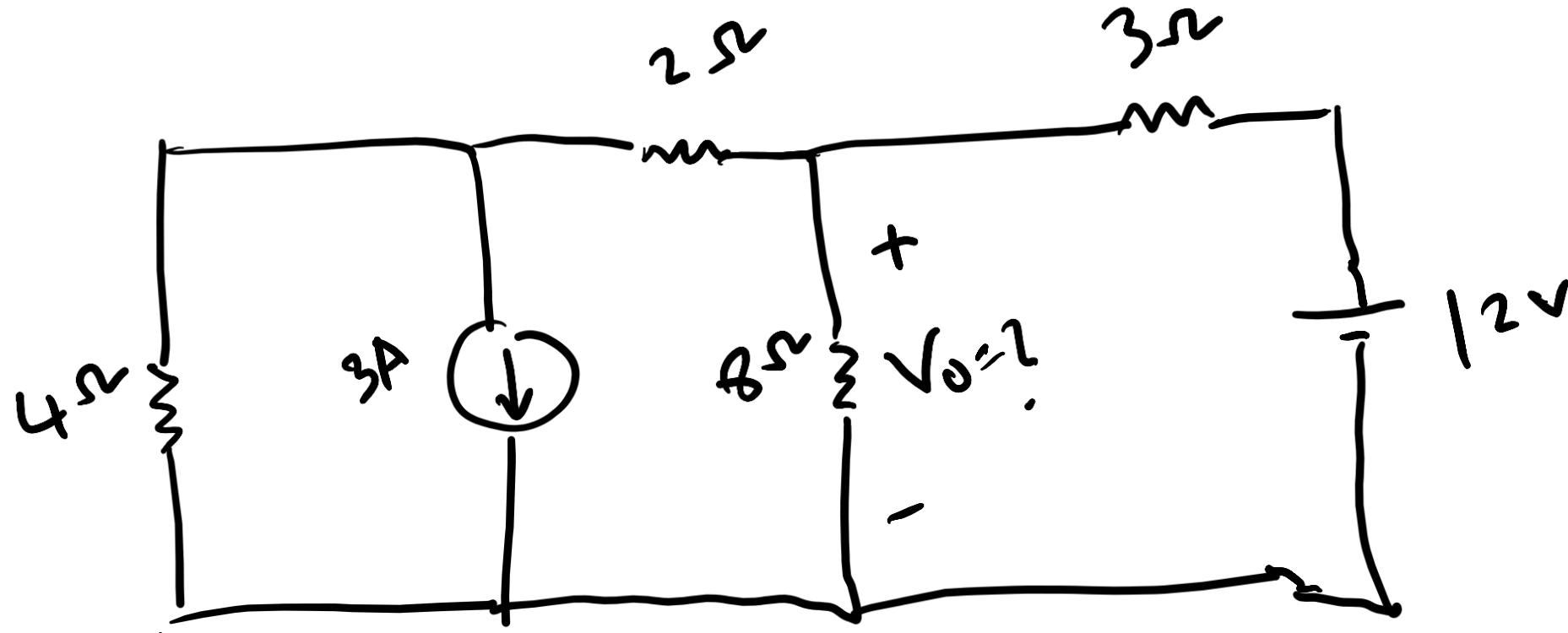
Source transformation – Voltage $\leftrightarrow$ Current



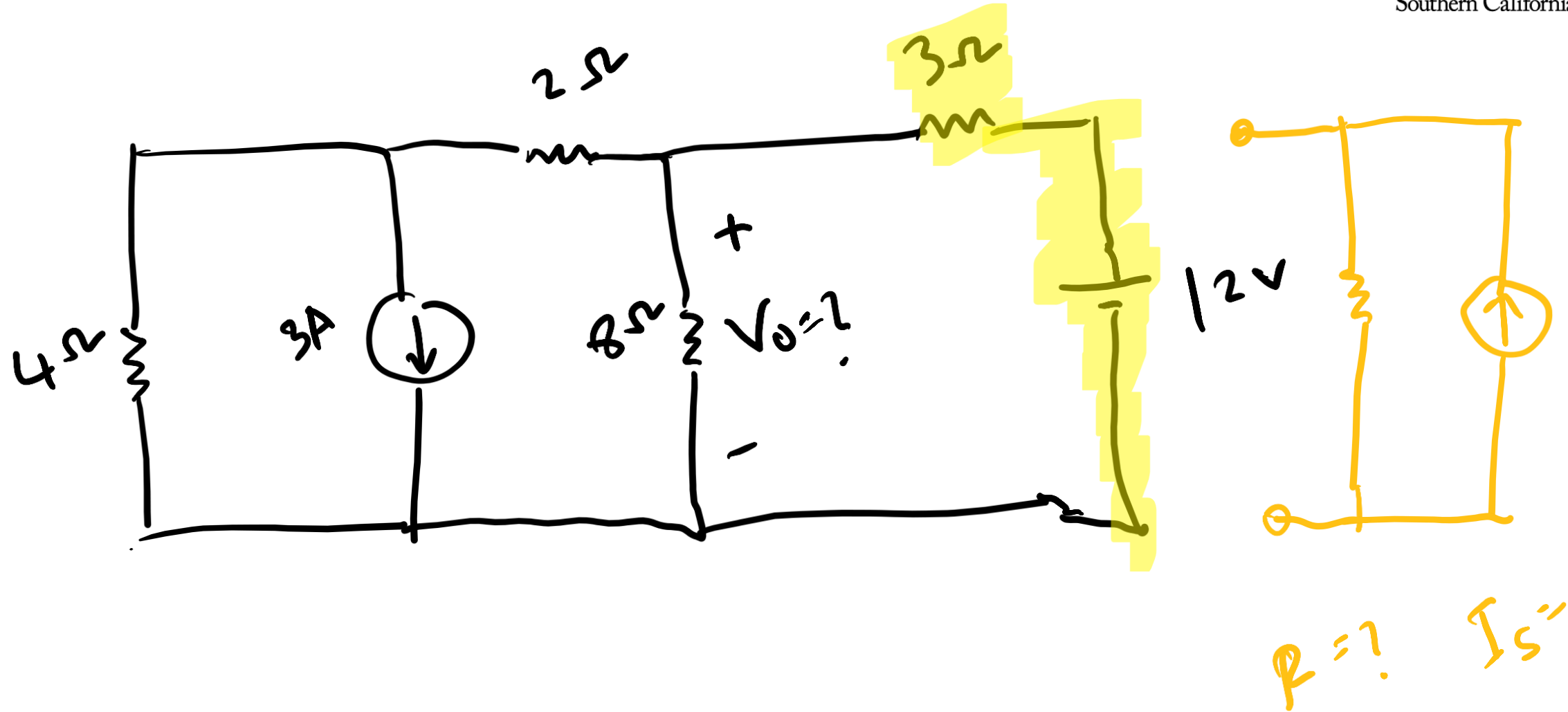
$$V_s = I_s R \Rightarrow$$

$$I_s = \frac{V_s}{R}$$

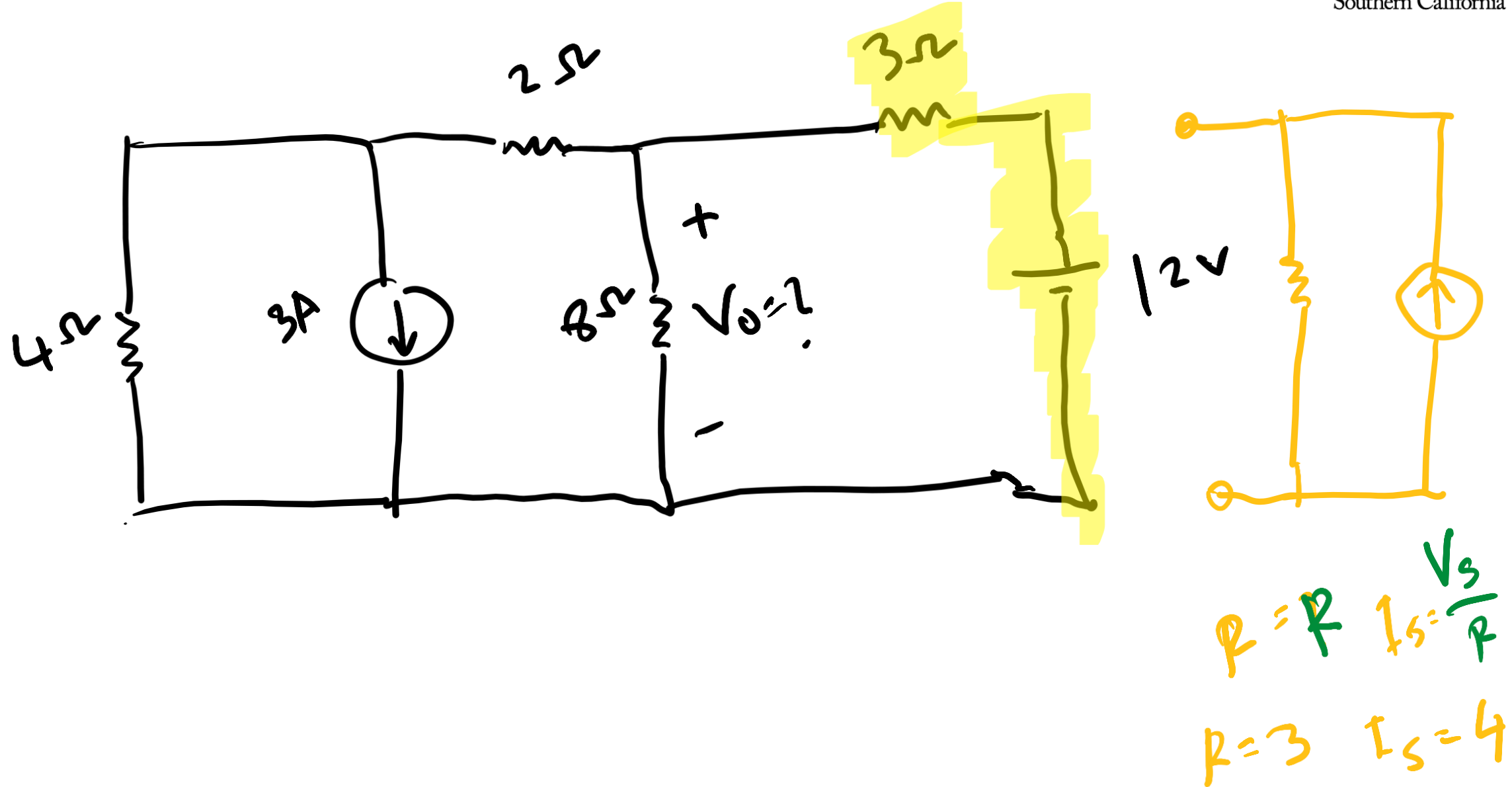
Voltage transformation example



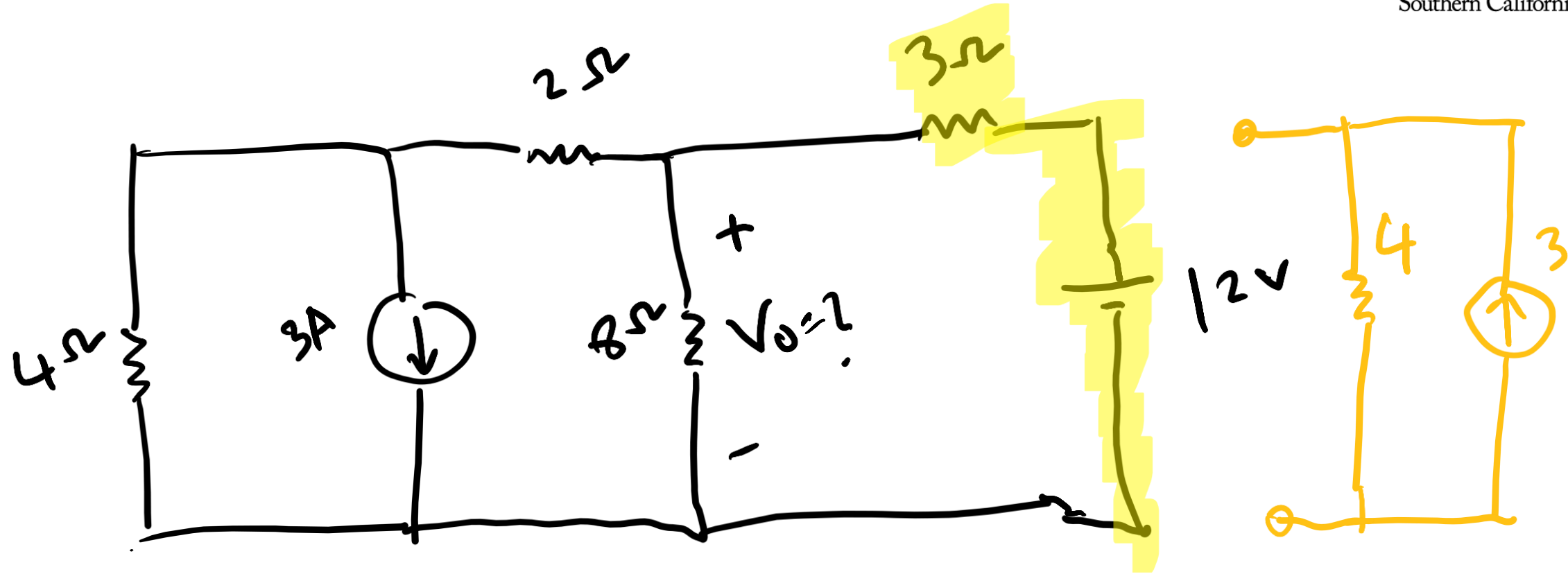
Voltage transformation example



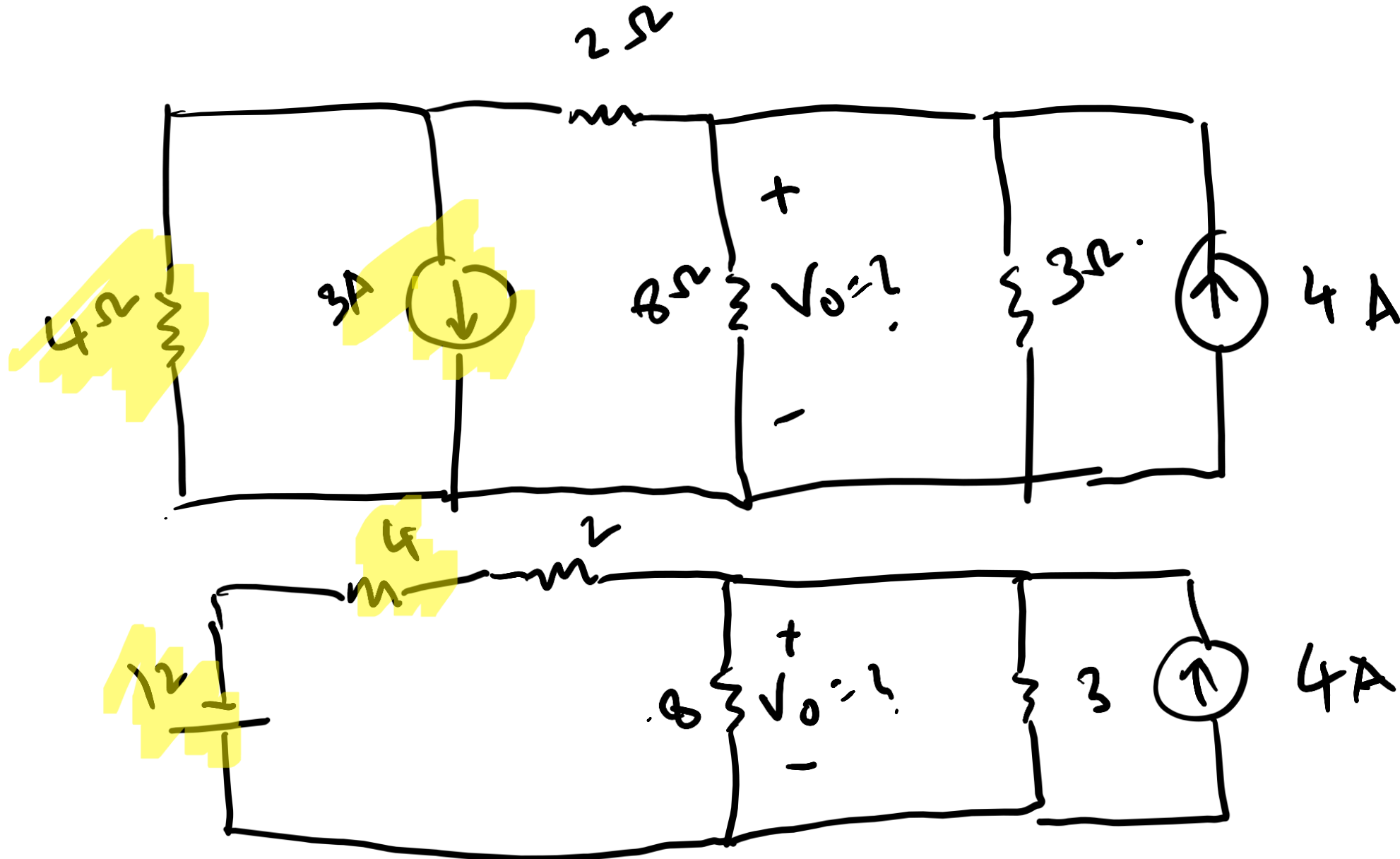
Voltage transformation example



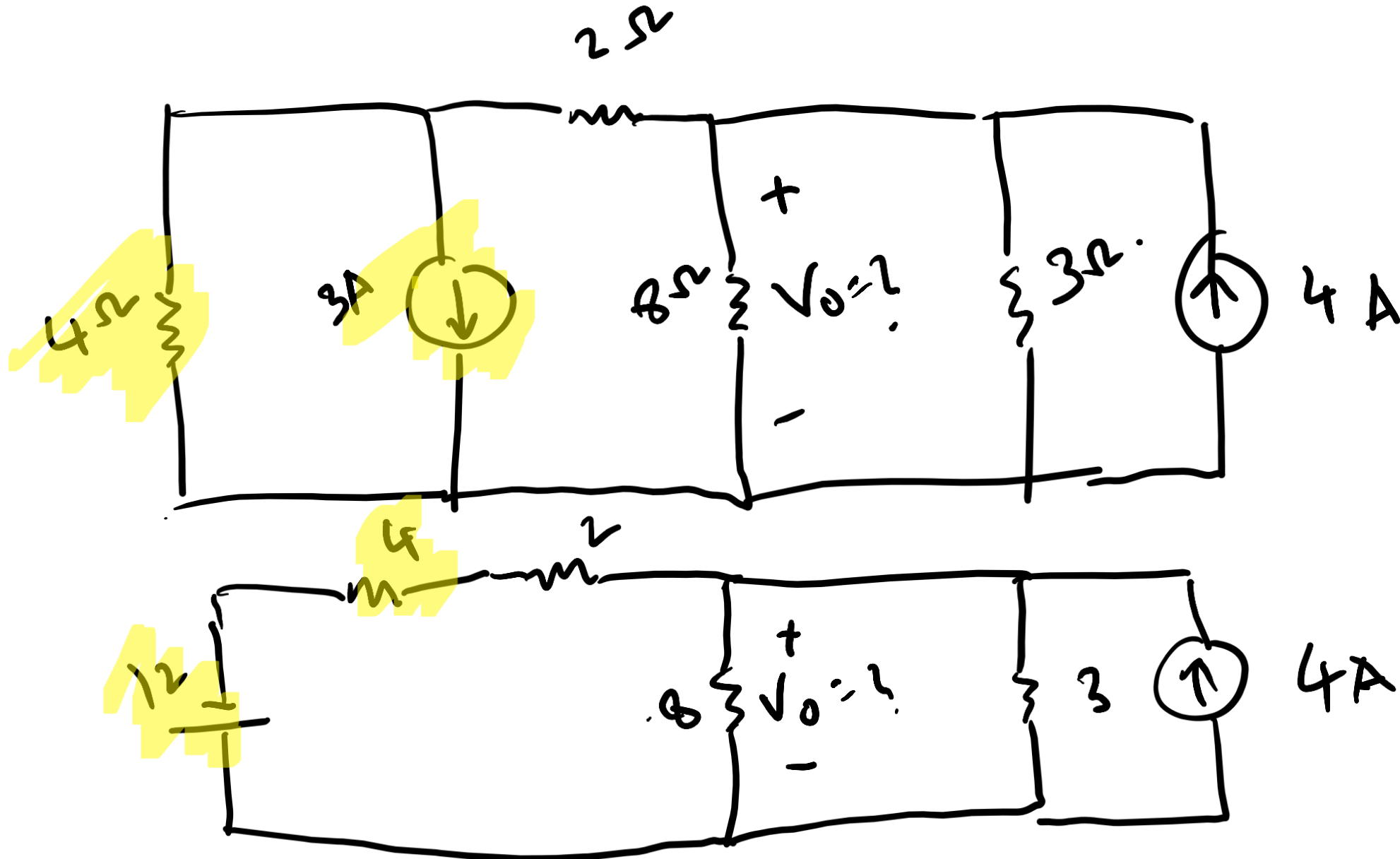
Voltage transformation example



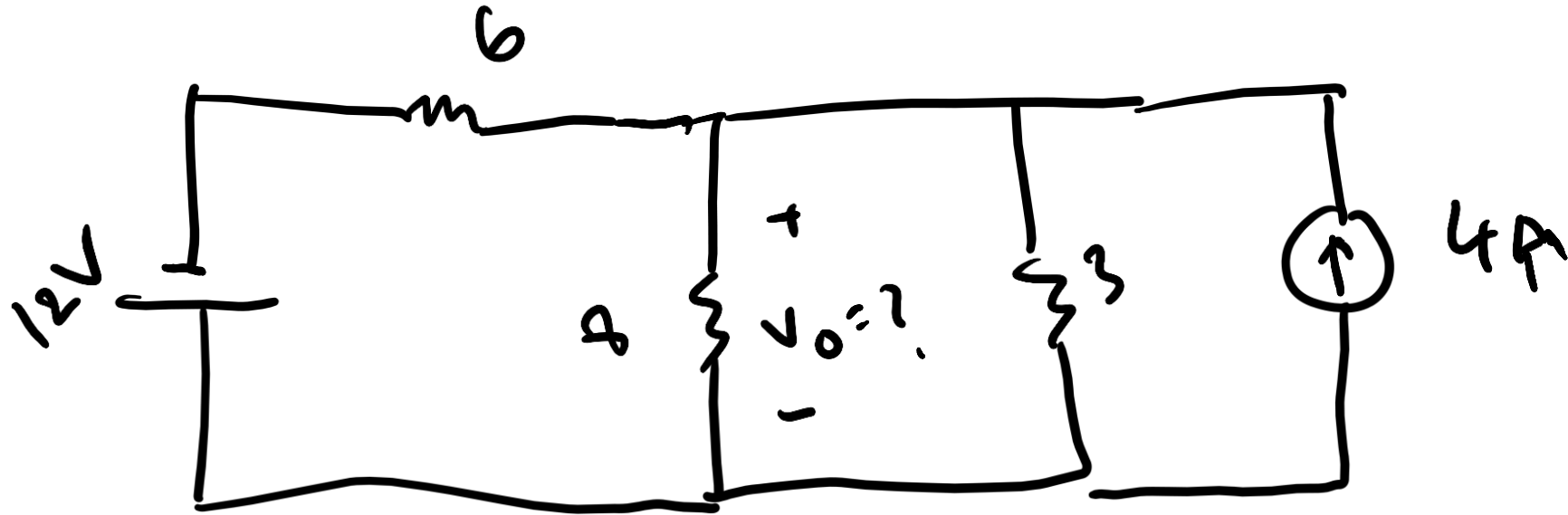
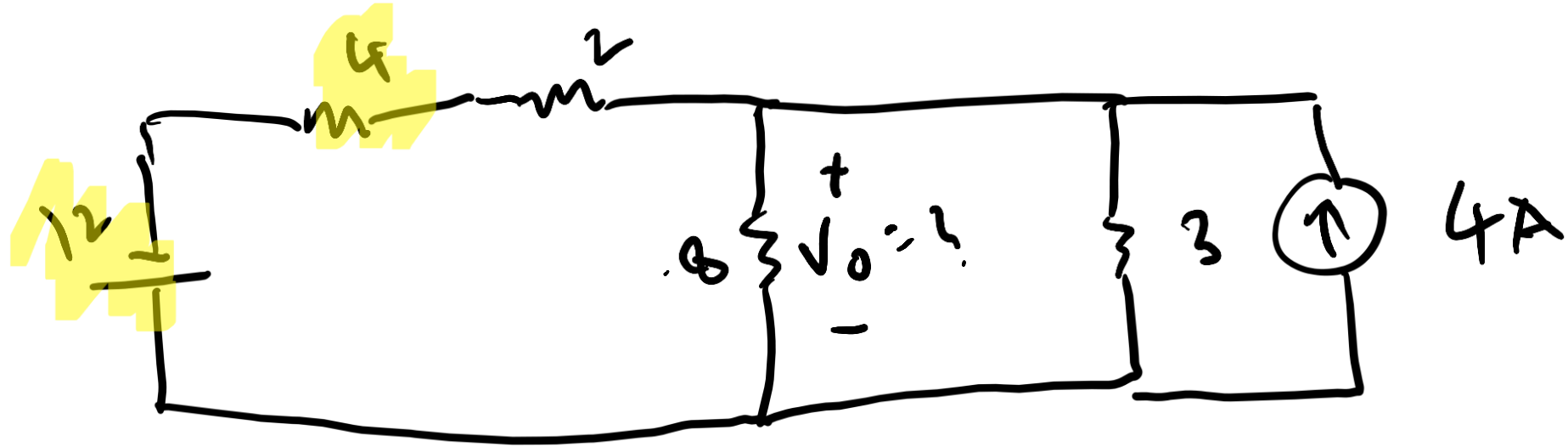
Voltage transformation example



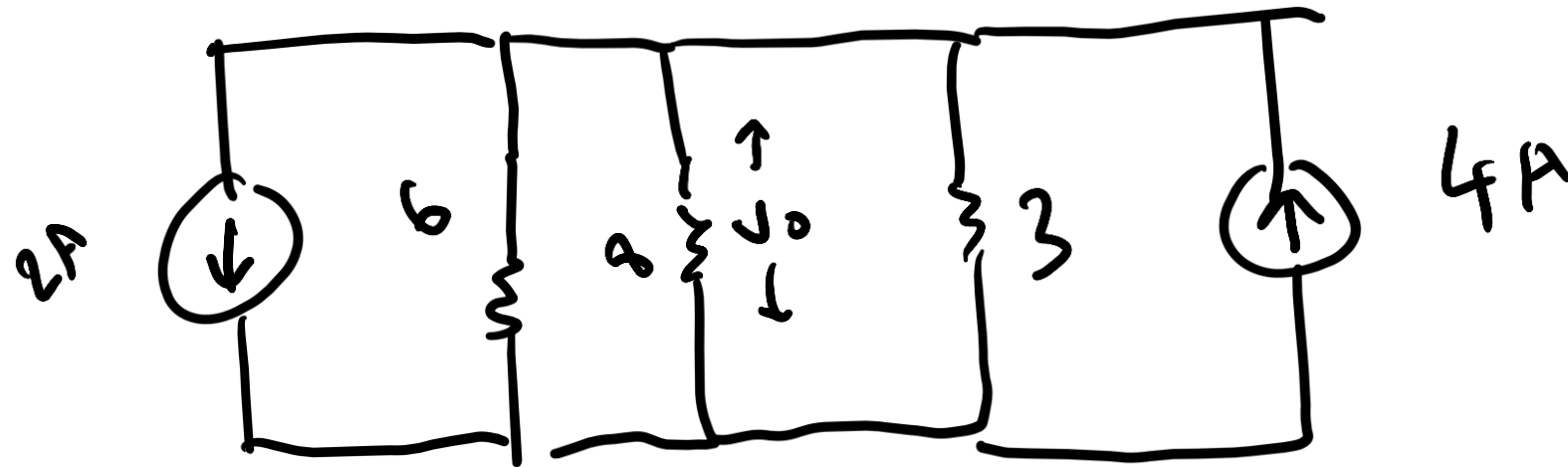
Voltage transformation example



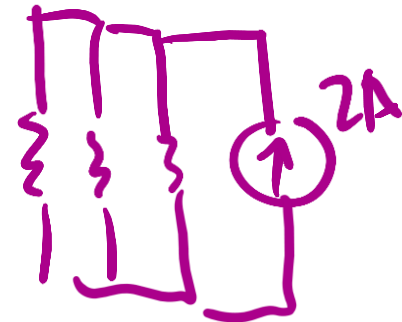
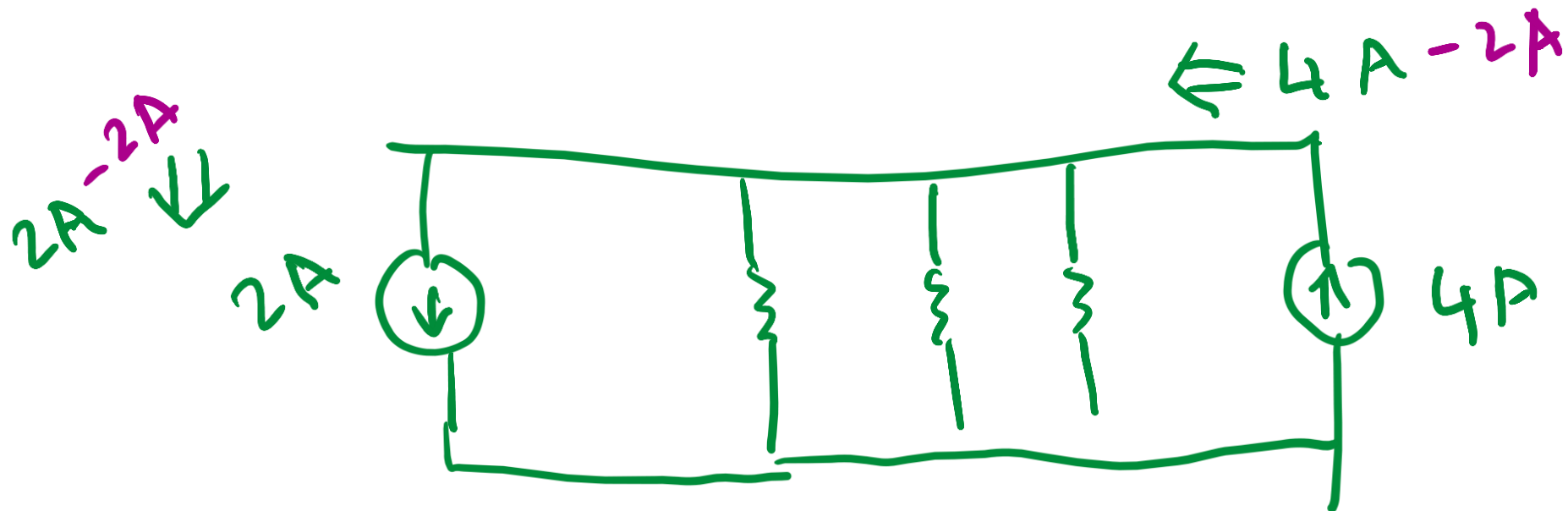
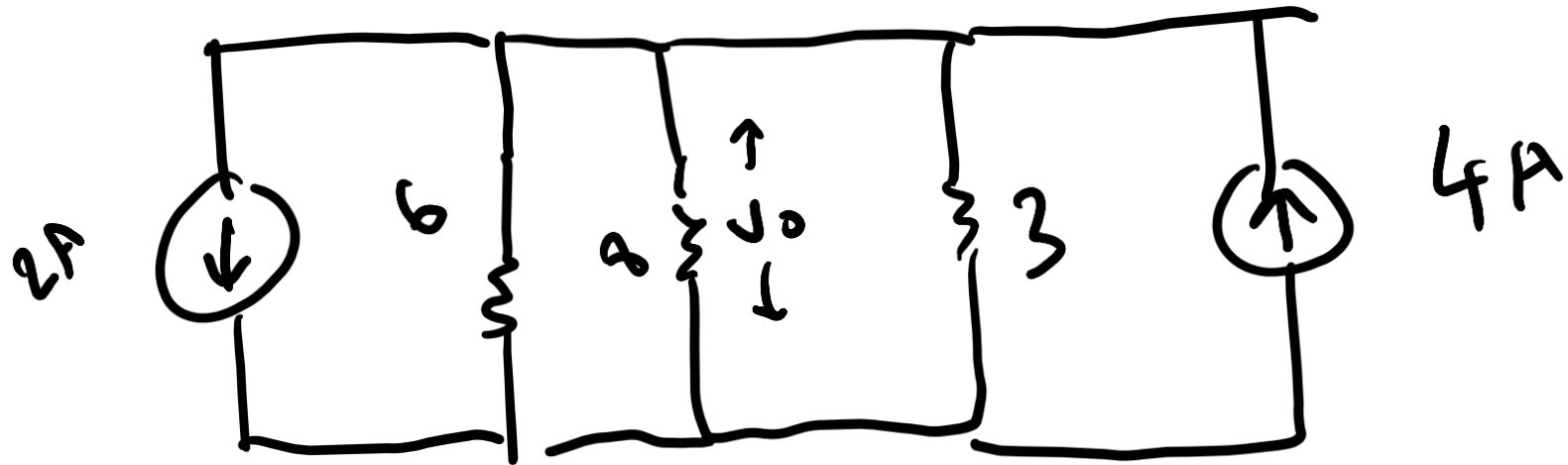
Voltage transformation example



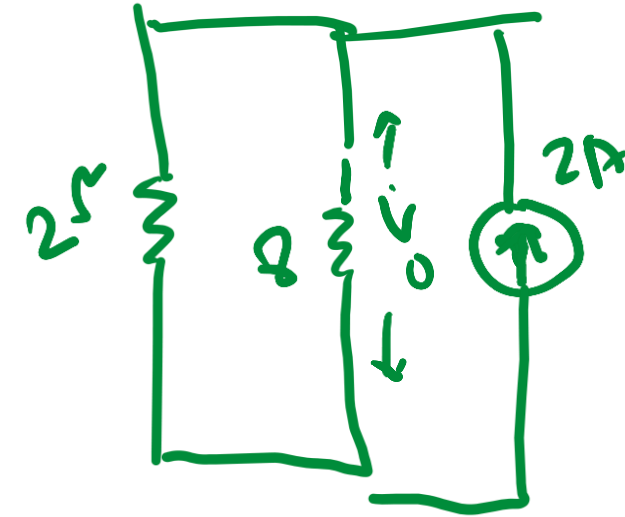
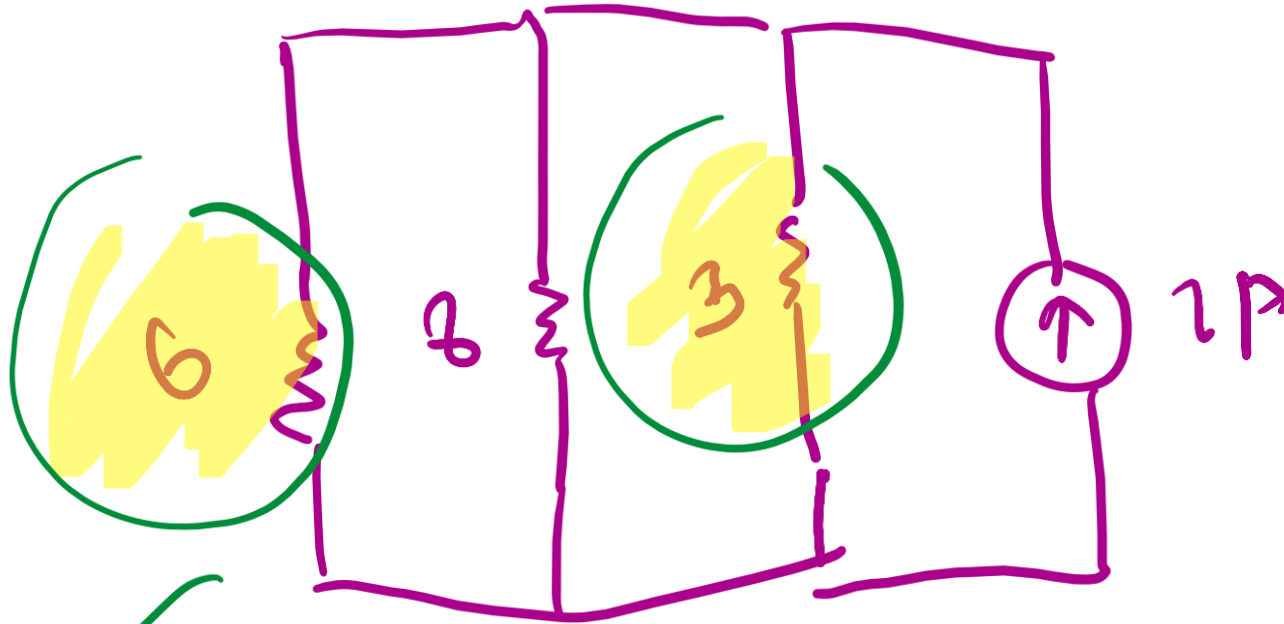
Voltage transformation example



Voltage transformation example

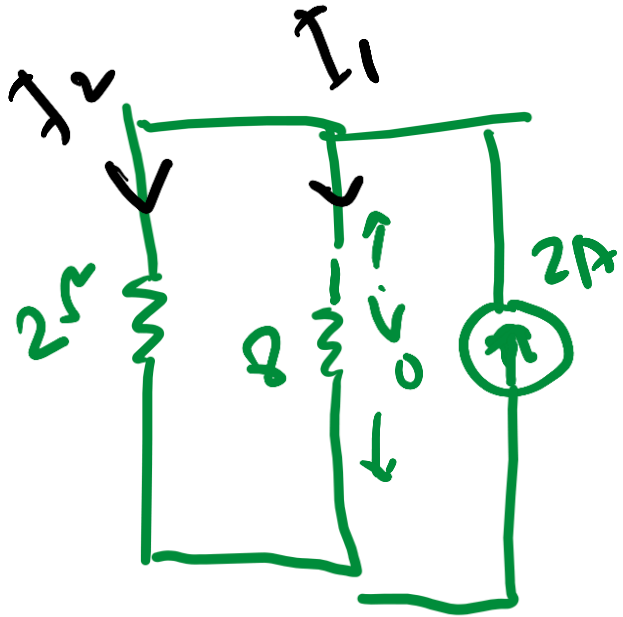


Voltage transformation example



$$\rightarrow \frac{1}{R_{eq}} = \frac{1}{6} + \frac{1}{3} = \frac{9}{18} \Rightarrow R_{eq} = 2$$

Voltage transformation example



$$I_1 = \frac{2 \times 2A}{10}$$

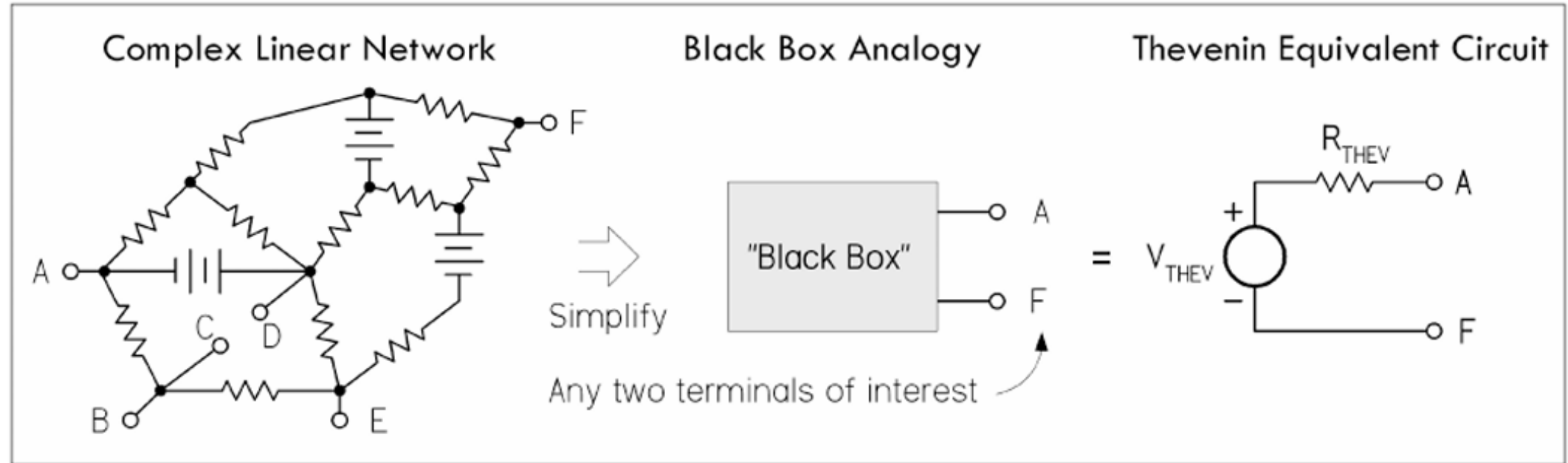
$$= .4A$$

$$V_0 = I_1 \cdot 8\Omega$$

$$= .4A \cdot 8\Omega$$

$$\boxed{= 3.2V}$$

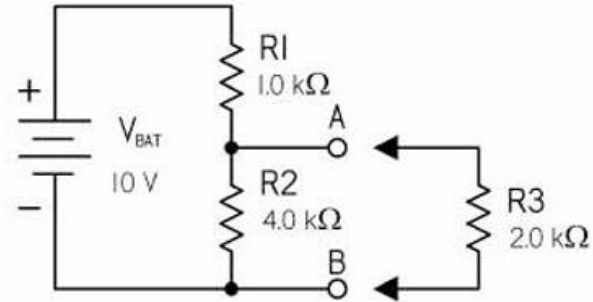
Equivalent circuits



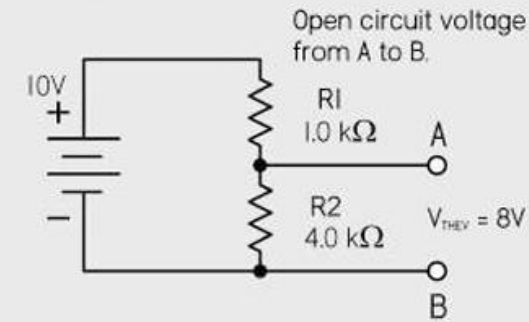
Thevenin's equivalent circuit

Thevenin's Theorem

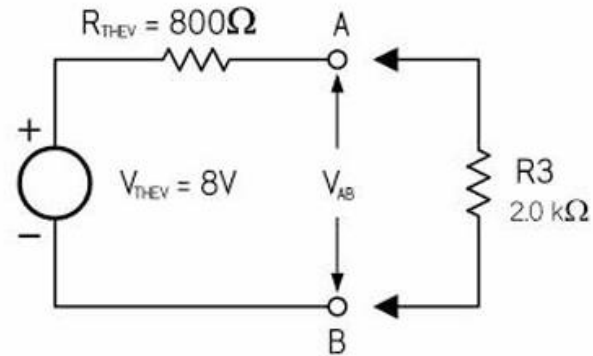
Real Circuit



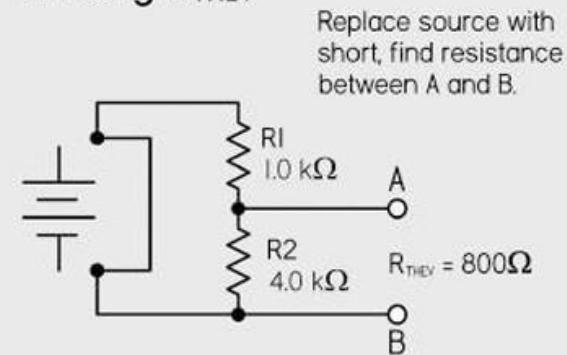
Finding V_{THEV}



Thevenin Equivalent Circuit



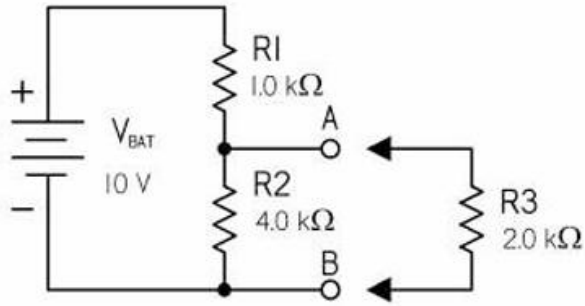
Finding R_{THEV}



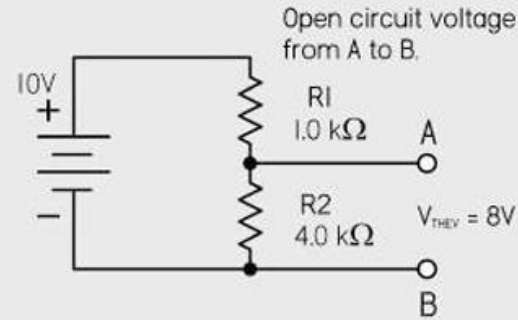
Thevenin's equivalent circuit

Thevenin's Theorem

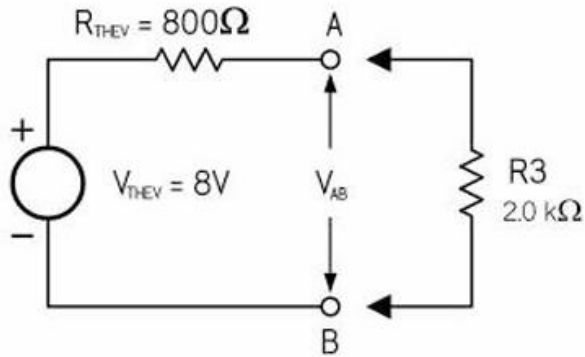
Real Circuit



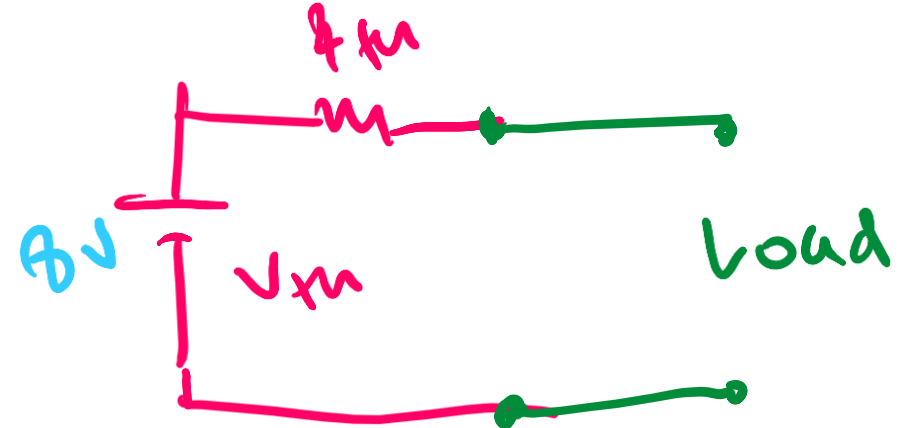
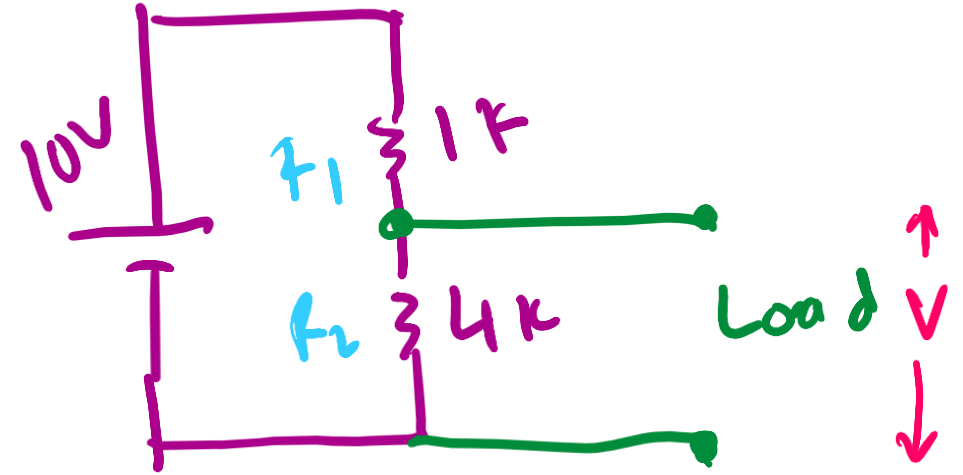
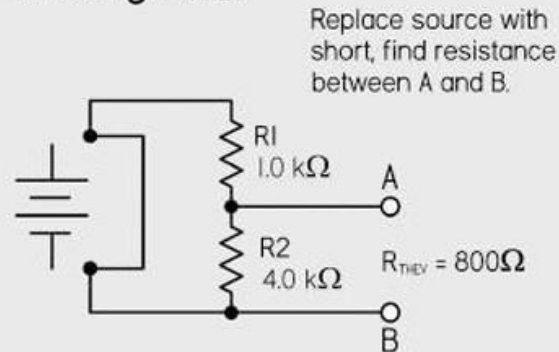
Finding V_{THEV}



Thevenin Equivalent Circuit

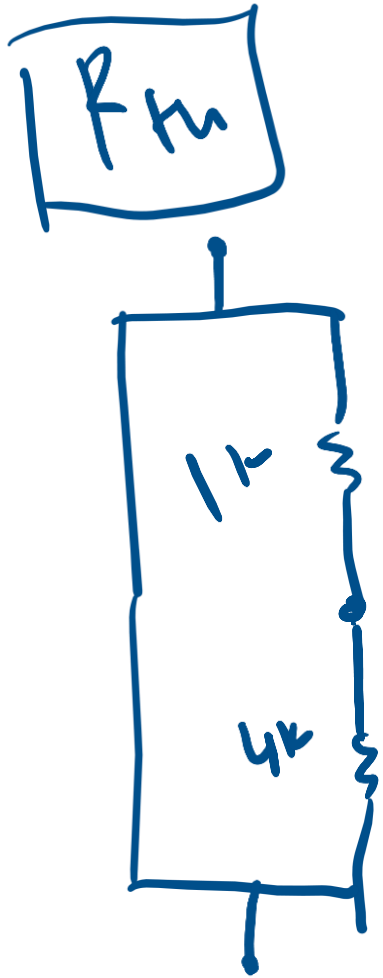


Finding R_{THEV}



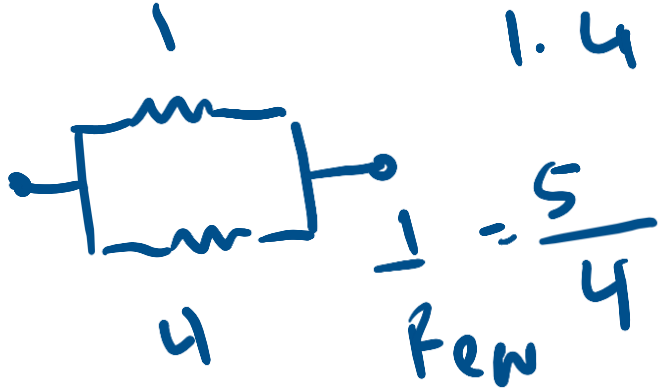
$$V_{THEV} = \frac{4 \cdot 10}{5} = 8V$$

Thevenin's equivalent circuit

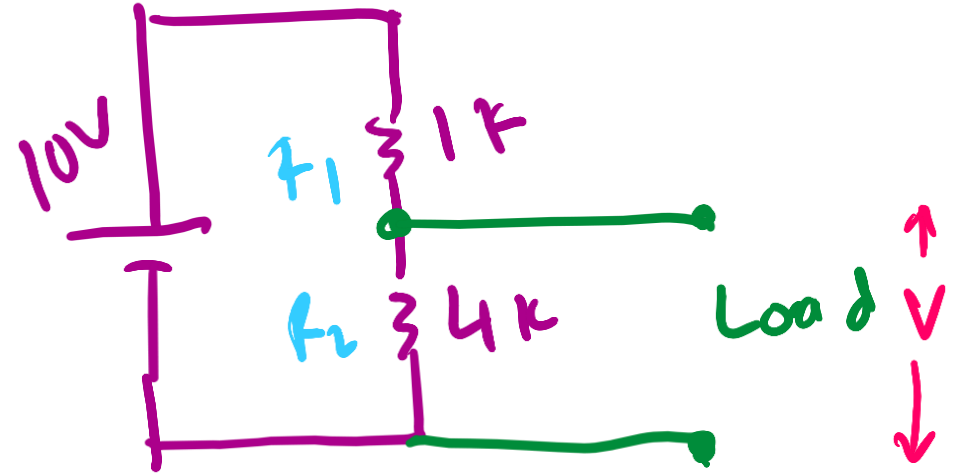


$$\frac{1}{R_{eq}} = \frac{1}{1k} + \frac{1}{4k}$$

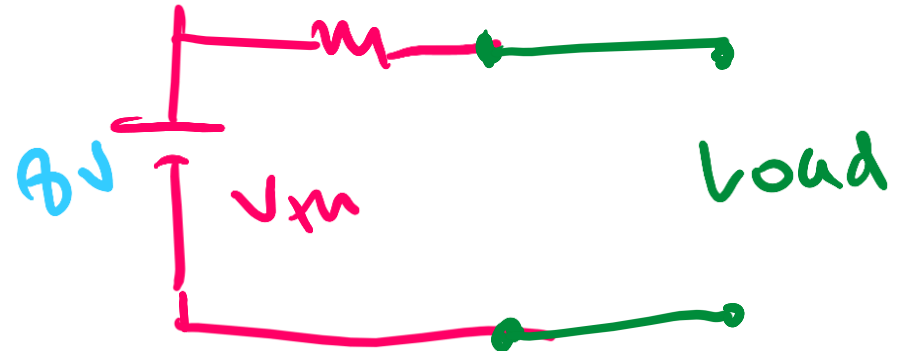
$$= \frac{5}{1.4}$$



$$R_{eq} = 800 \Omega$$



$$R_{eq} = 800 \Omega$$

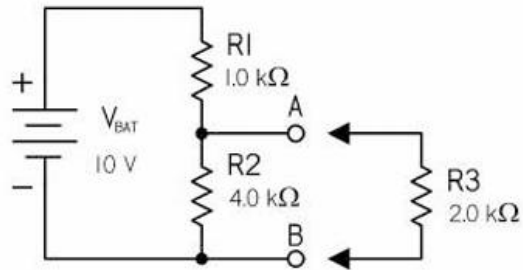


$$V_{OC} = \frac{4 \cdot 10}{5} V = 8V$$

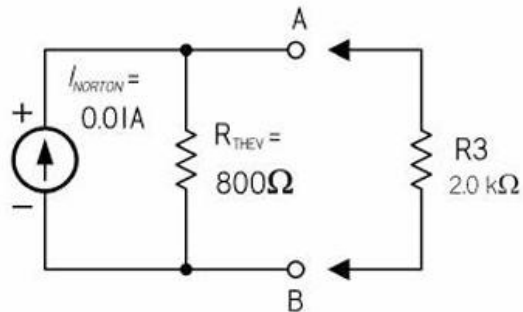
Norton's equivalent circuit

Norton's Theorem

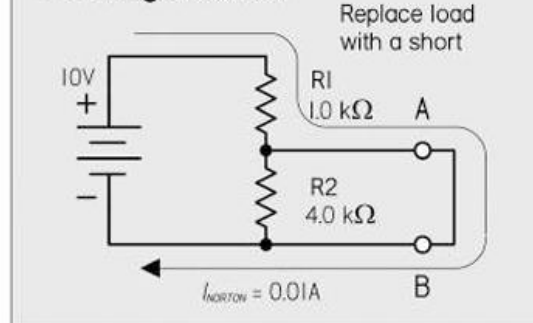
Real Circuit



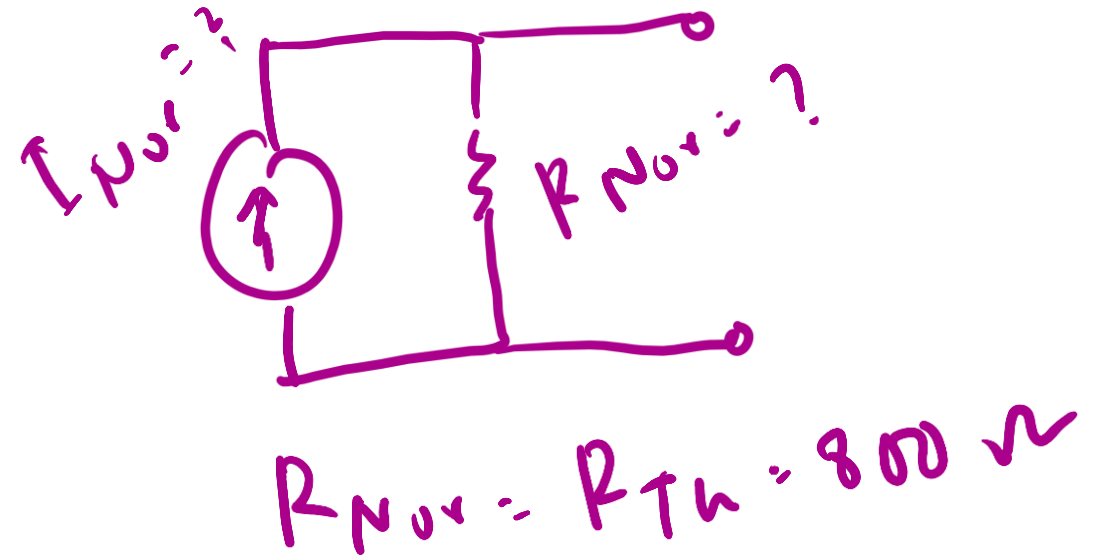
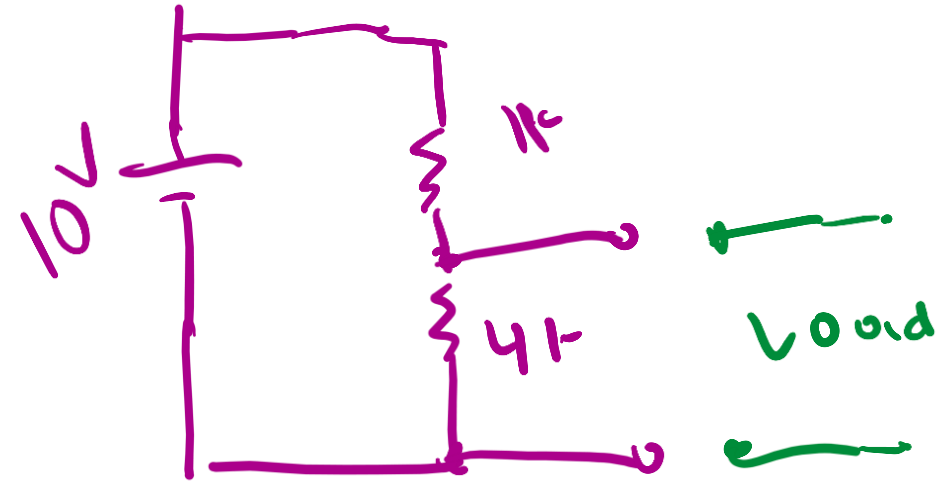
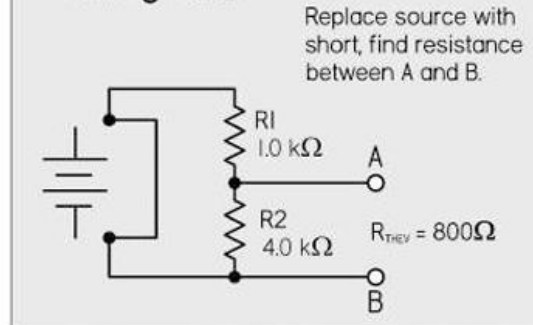
Norton Equivalent Circuit



Finding I_{NORTON}

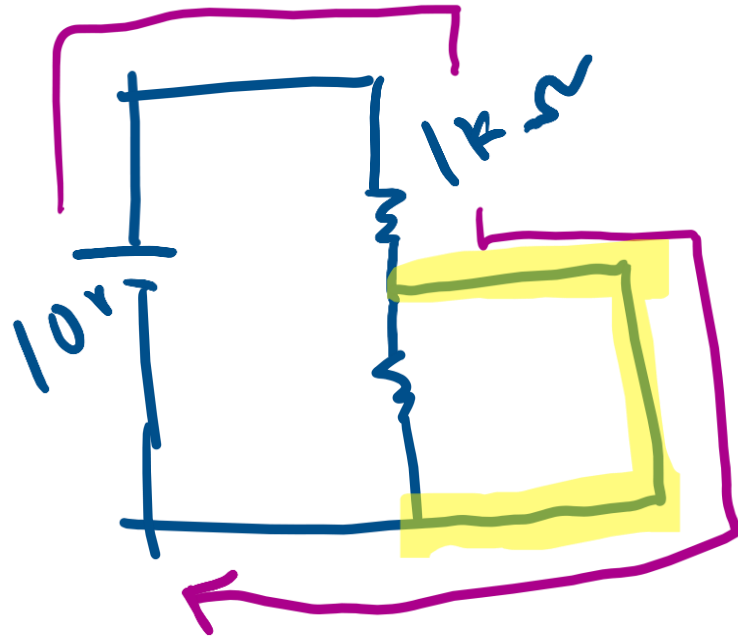


Finding R_{THEV}



Norton's equivalent circuit

Replace load with a
Short.

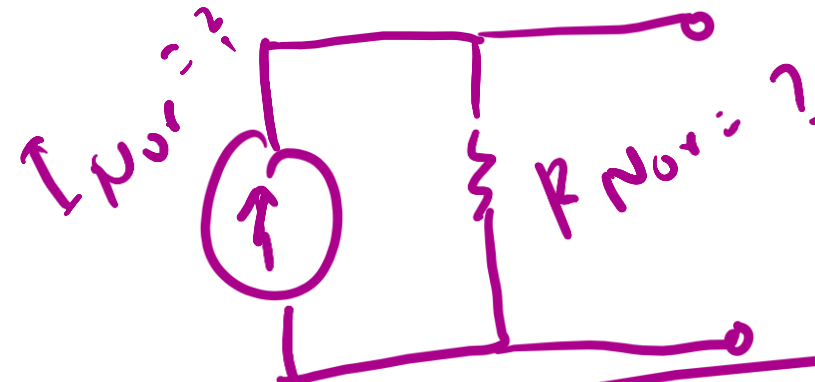
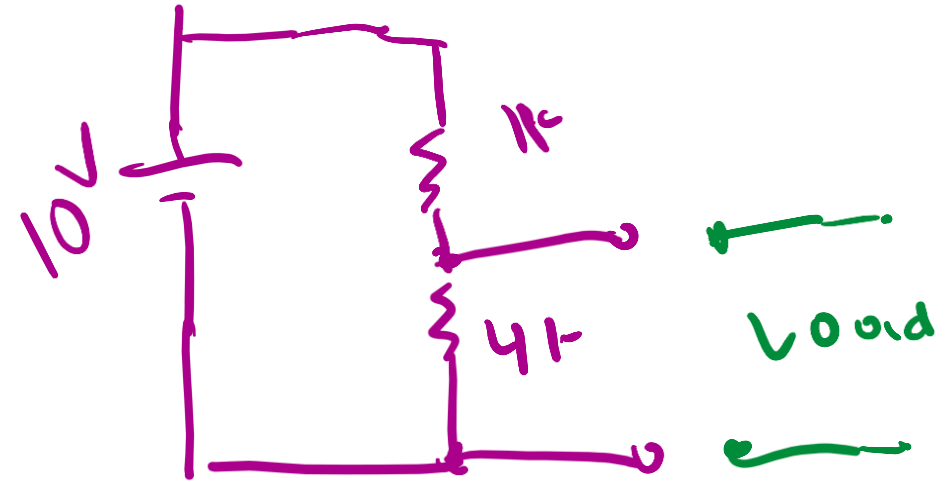


$$V = IR$$

$$I = \frac{V}{R} = \frac{10V}{1k}$$

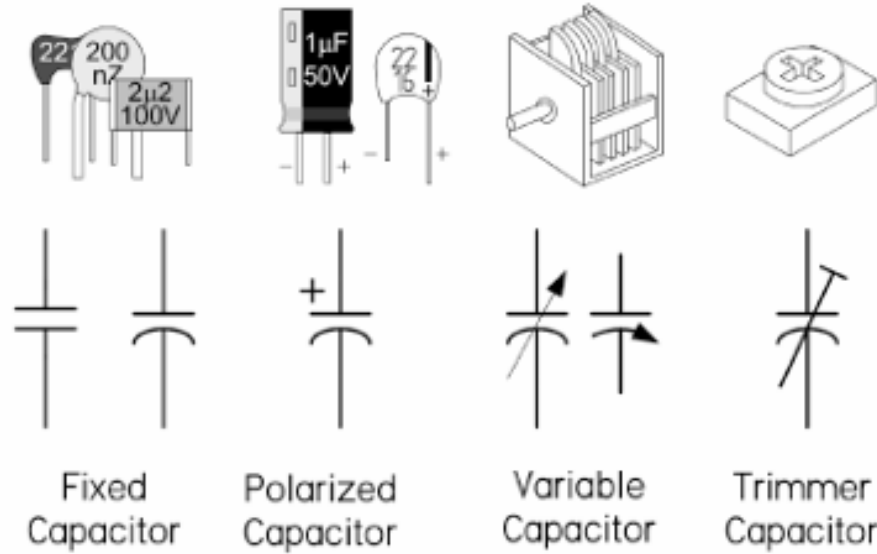
$$= 0.01A$$

$$I_{NOR} = 0.01A$$

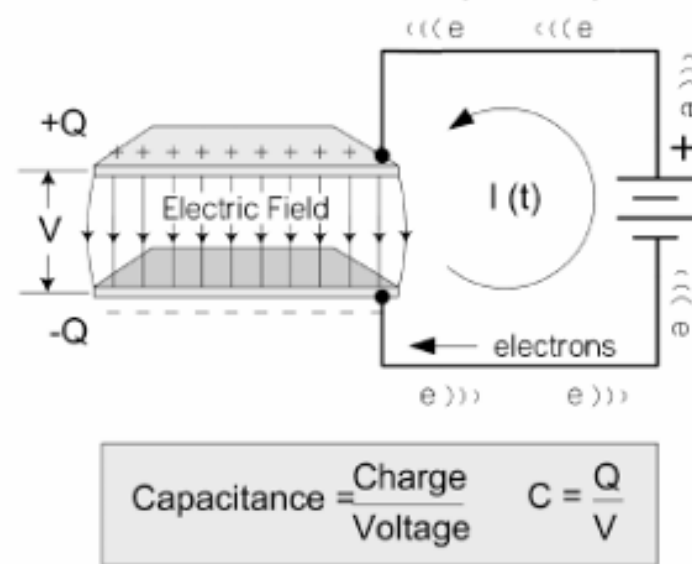


$$R_{NOR} = R_{TH} = 800\Omega$$

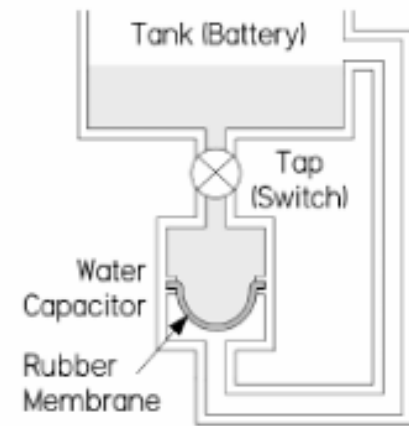
Capacitor Symbols



Parallel Plate Capacitor

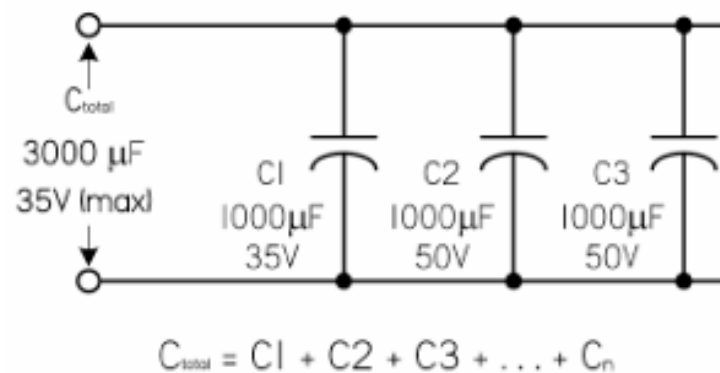


Water Analogy



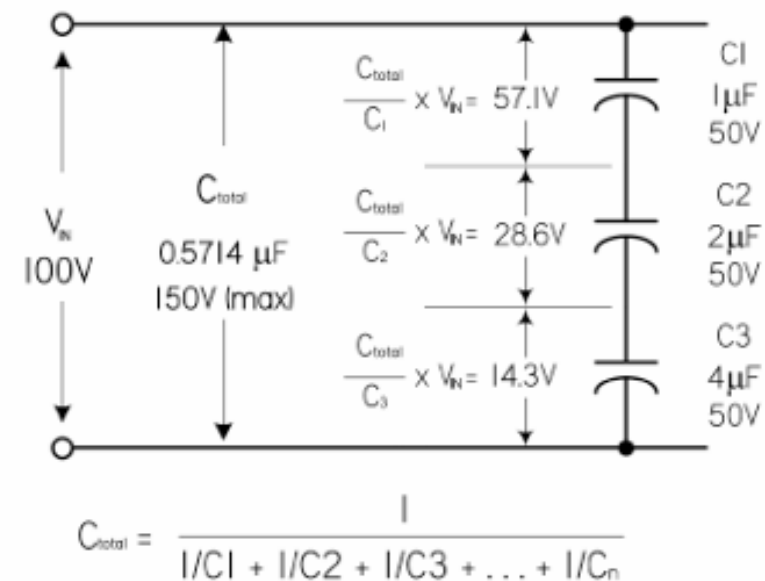
Capacitors In Parallel

Increases the total capacitance, but limits max. voltage rating to that of smallest rated capacitor.



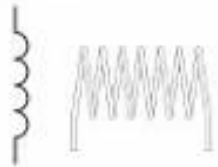
Capacitors In Series

Increases max voltage rating, but decreases capacitance.

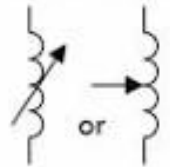


Inductor Symbols

Air Core



Adjustable



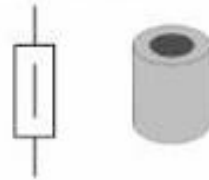
non magnetic
slug^c



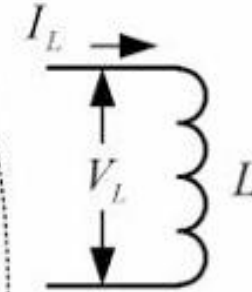
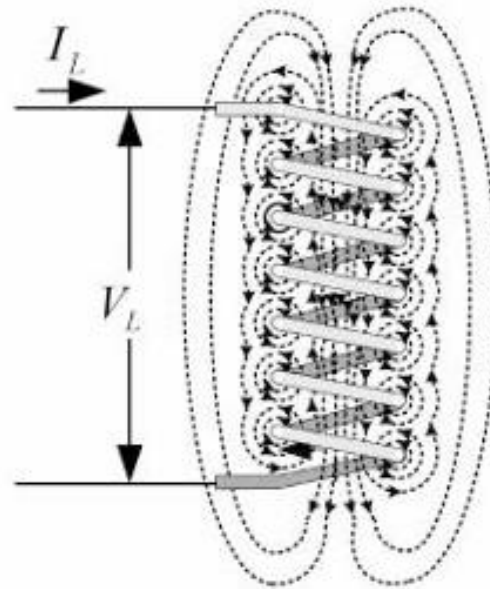
Magnetic or Iron Core



Ferrite Bead



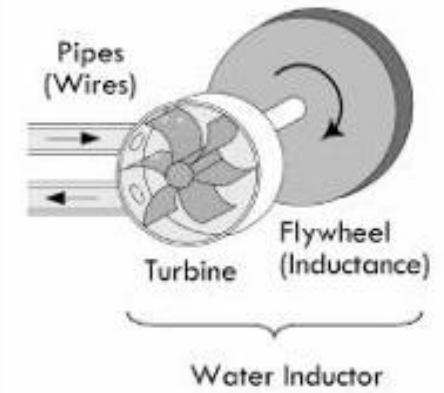
Air-Core Inductor



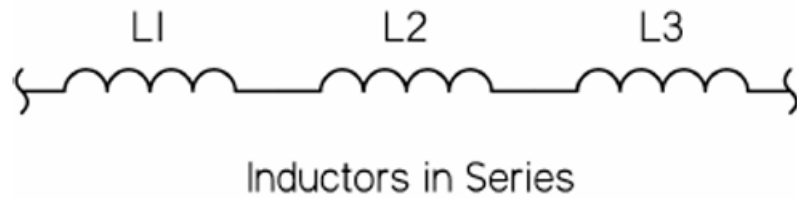
$$V_L = L \frac{dI_L}{dt}$$

$$I_L = \frac{1}{L} \int V_L dt$$

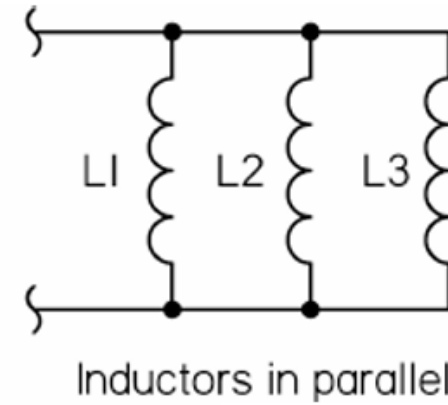
Water Analogy



Inductors

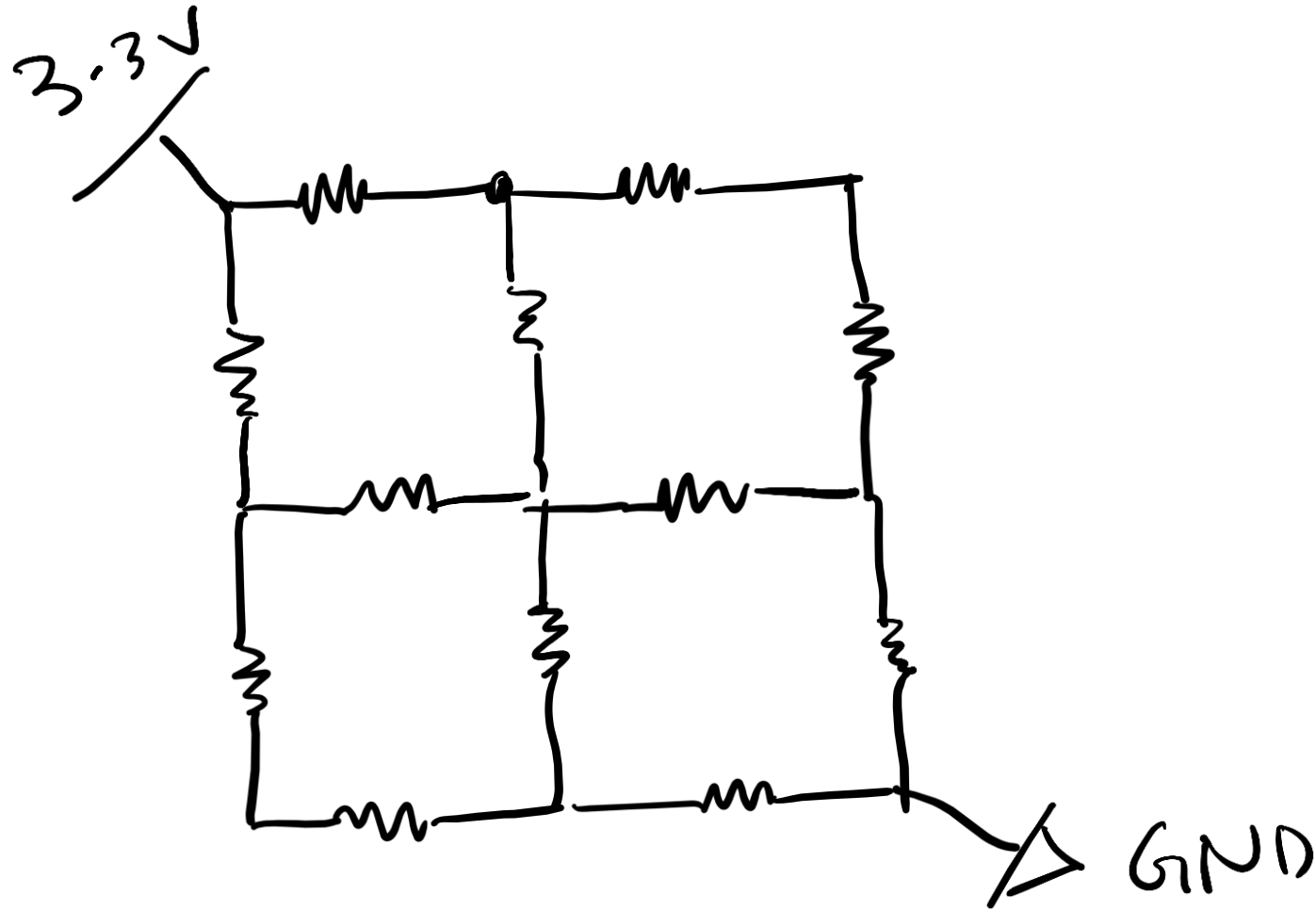


$$L_{\text{eq}} = L_1 + L_2 + L_3$$

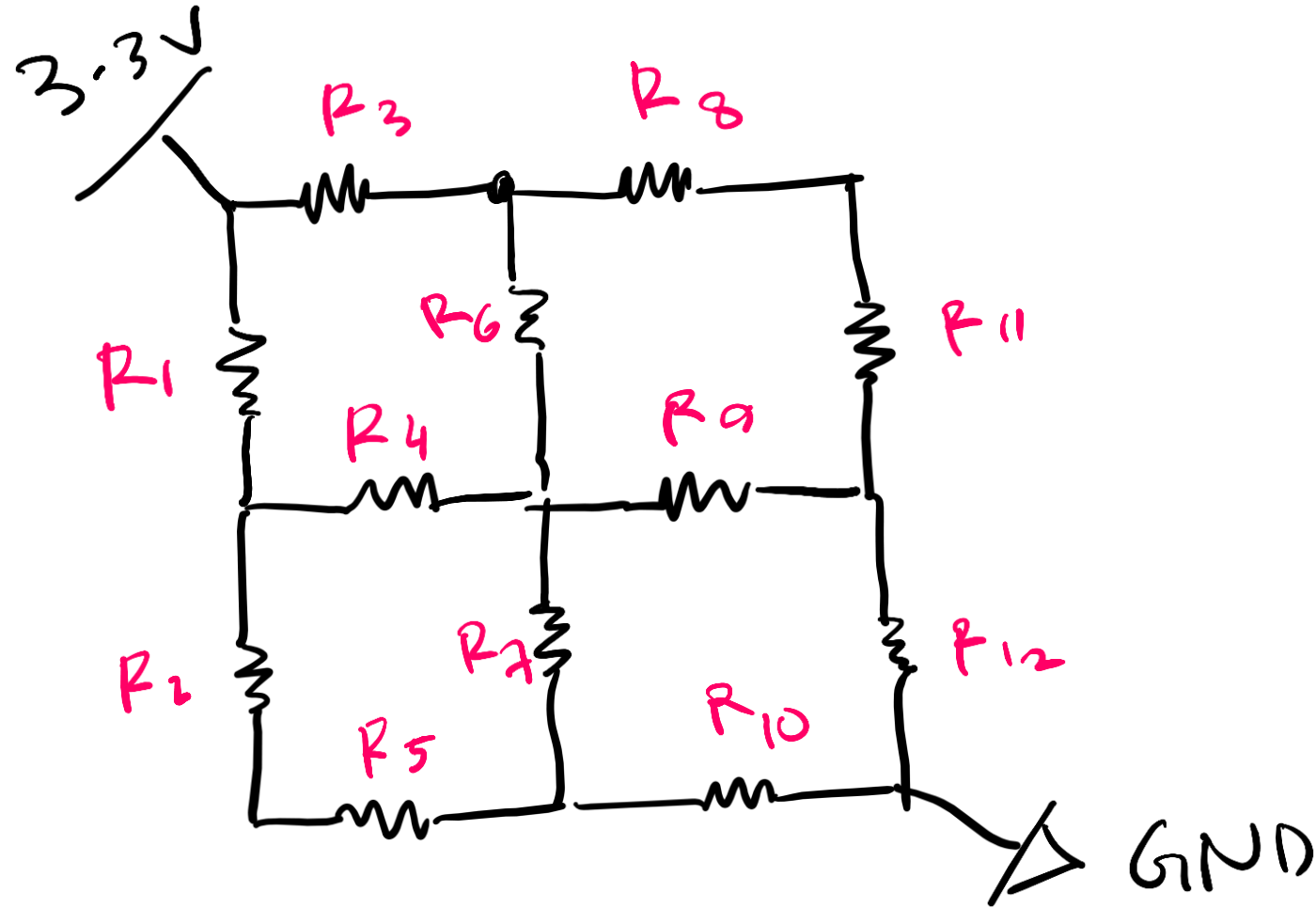


$$\frac{1}{L_{\text{eq}}} = \frac{1}{L_1} + \frac{1}{L_2} + \frac{1}{L_3}$$

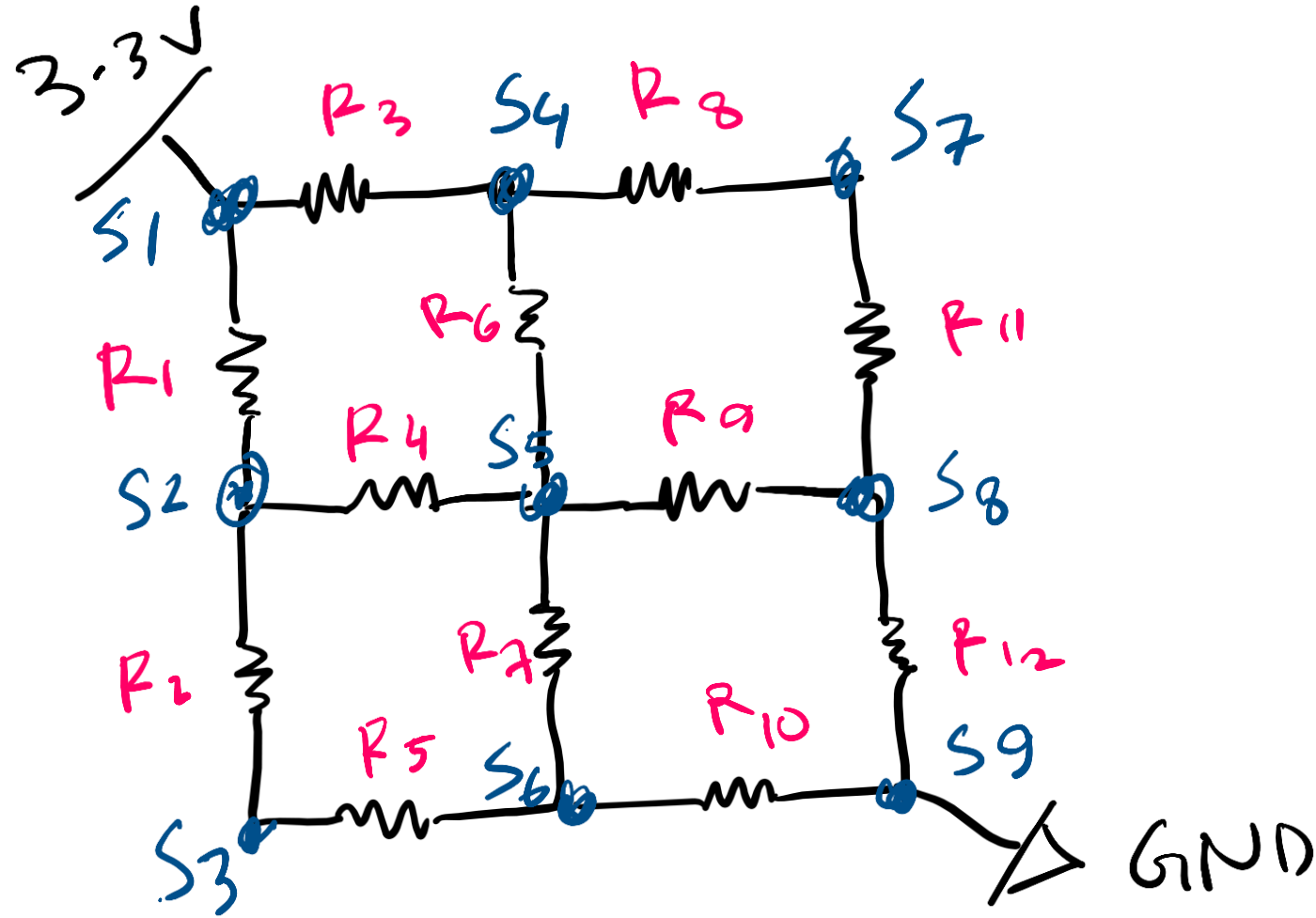
Touchpad – Network 2 in the demo board



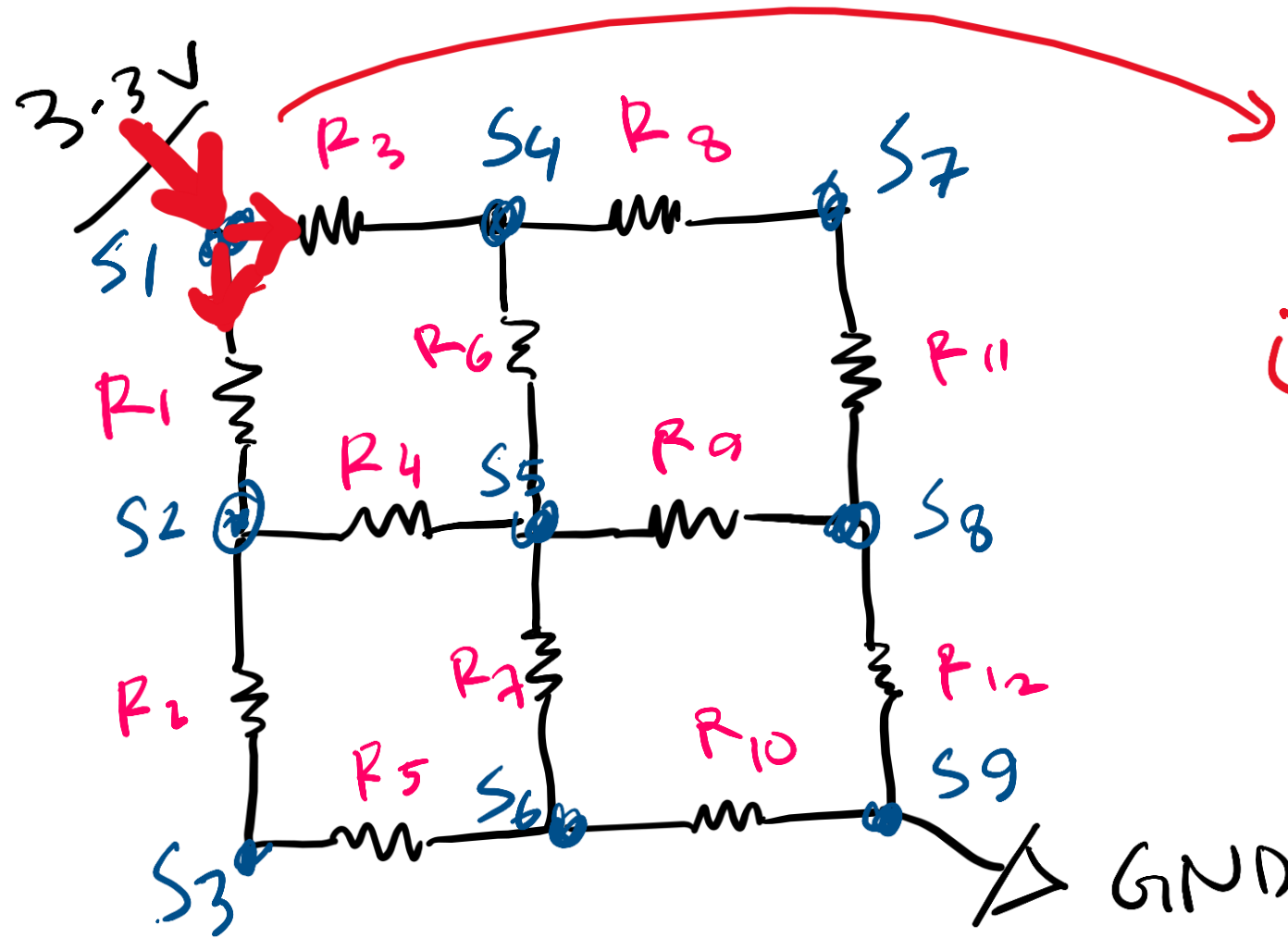
Touchpad – Network 2 in the demo board



Touchpad – Network 2 in the demo board



Touchpad – Network 2 in the demo board



$$i_{tot} = i_{R_1} + i_{R_3}$$

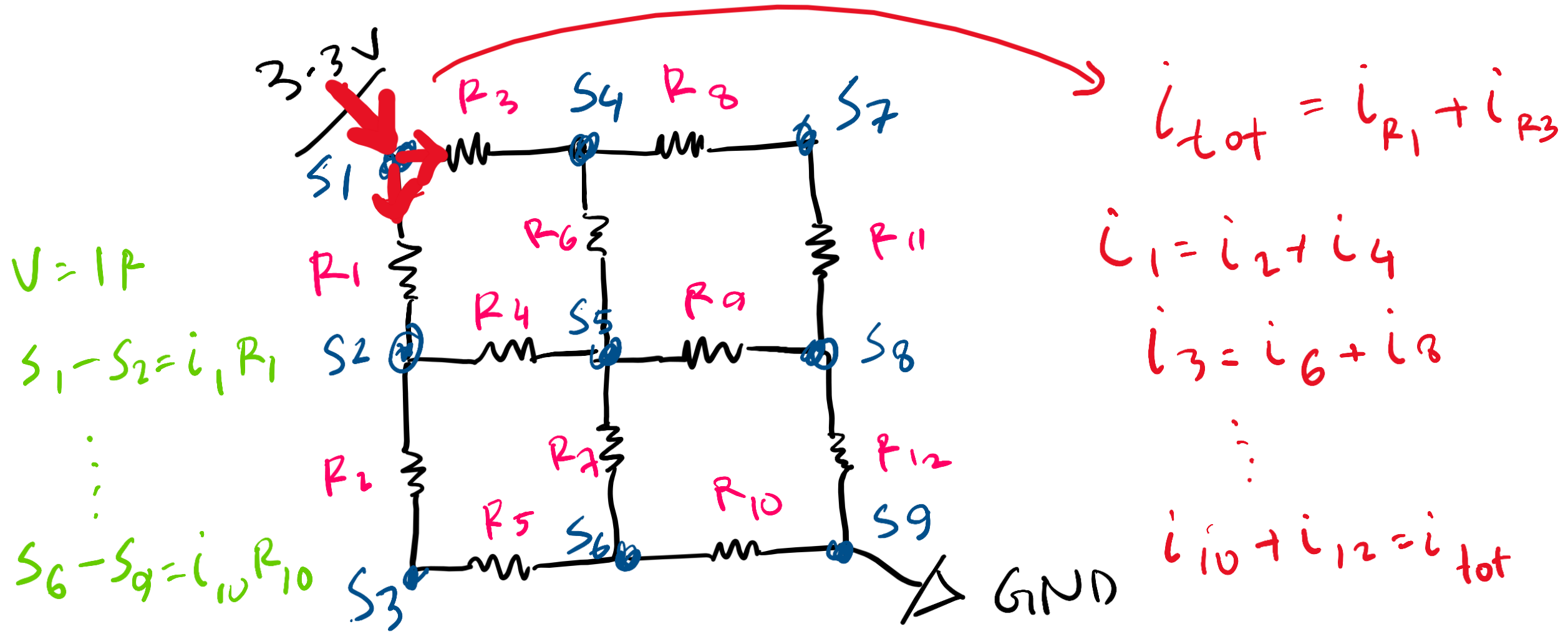
$$i_1 = i_2 + i_4$$

$$i_3 = i_6 + i_8$$

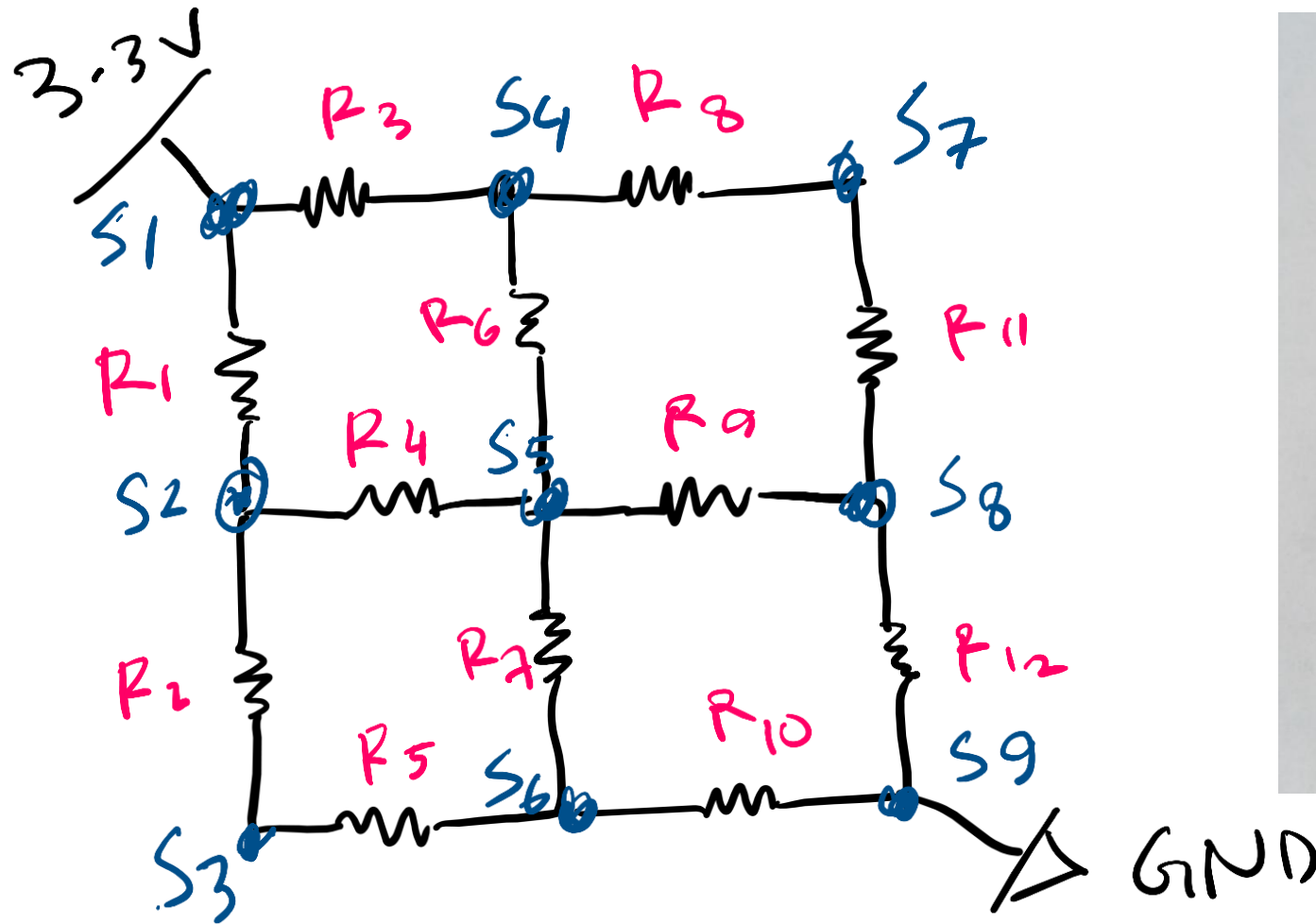
$\vdots$

$$i_{10} + i_{12} = i_{tot}$$

Touchpad – Network 2 in the demo board



Touchpad – Network 2 in the demo board



$$\begin{aligned}
 I_t &= i_{R1} + i_{R2} \\
 i_{R1} &= i_{R2} + i_{R4} \\
 i_{R3} &= i_{R6} + i_{R8} \\
 i_{R7} &= i_{R5} \\
 i_{R8} &= i_{R11} \\
 i_{R4} + i_{R6} &= i_{R7} + i_{R9} \\
 i_{R5} + i_{R7} &= i_{R10} \\
 i_{R9} + i_{R11} &= i_{R12} \\
 i_{R10} + i_{R12} &= I_t
 \end{aligned}$$

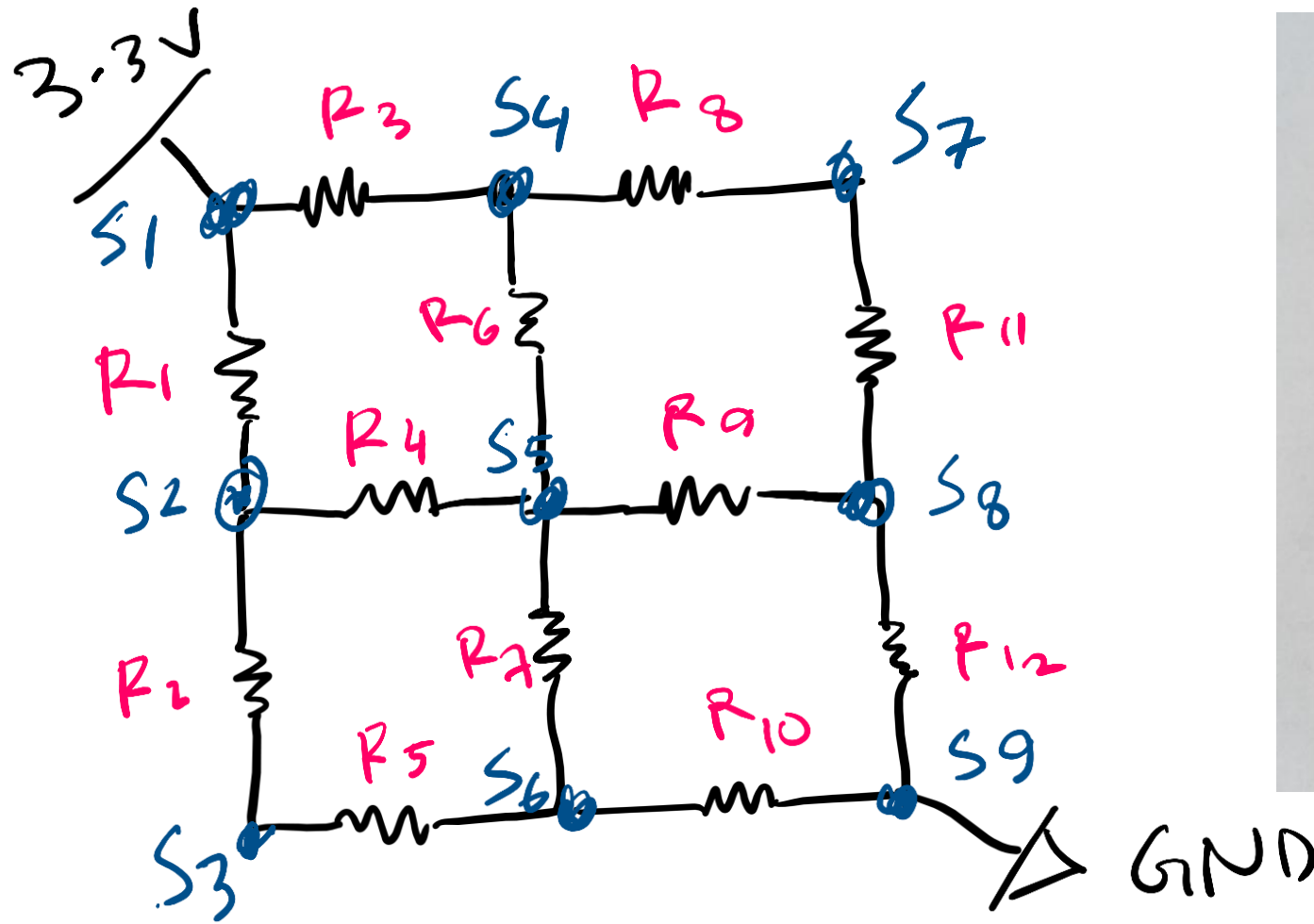
$$\begin{aligned}
 V_{R1} &= S_1 - S_2 = i_1 R_1 = S_1 - S_2 \\
 V_{R2} &= i_2 R_2 = S_2 - S_3 \\
 V_{R3} &= i_3 R_3 = S_1 - S_4 \\
 V_{R4} &= i_4 R_4 = S_2 - S_5 \\
 V_{R5} &= i_5 R_5 = S_3 - S_6 \\
 i_{R6} &= S_4 - S_5 \\
 i_{R7} &= S_5 - S_6 \\
 i_{R8} &= S_4 - S_7 \\
 i_{R9} &= S_5 - S_8 \\
 i_{R10} &= S_6 - S_9 \\
 i_{R11} &= S_7 - S_8 \\
 i_{R12} &= S_8 - S_9
 \end{aligned}$$

gen

gen

$Ax = y$

Touchpad – Network 2 in the demo board

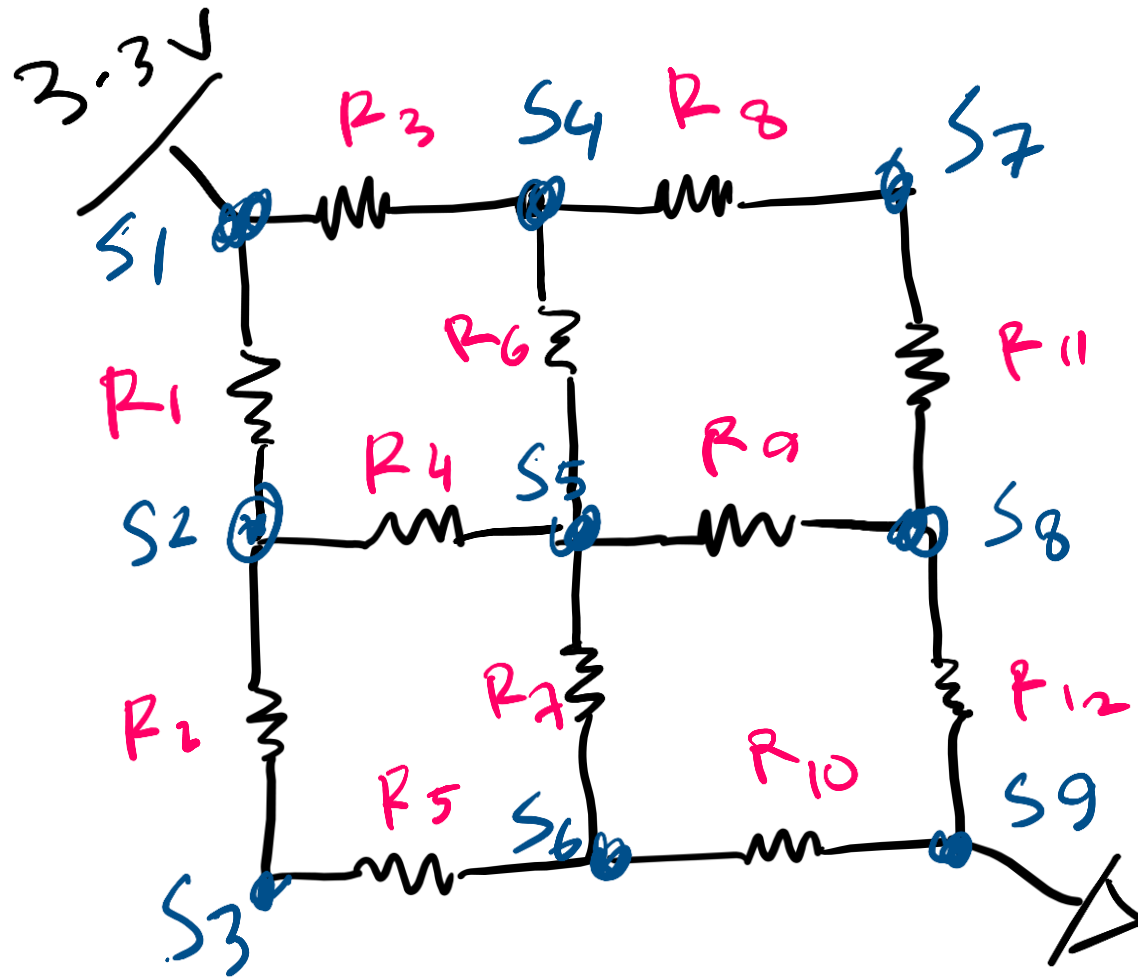


$$\begin{aligned}
 i_t &= i_{R1} + i_{R2} \\
 i_{R1} &= i_{R2} + i_{R4} \\
 i_{R3} &= i_{R6} + i_{R8} \\
 i_{R2} &= i_{R5} \\
 i_{R8} &= i_{R11} \\
 i_{R4} + i_{R6} &= i_{R7} + i_{R9} \\
 i_{R5} + i_{R7} &= i_{R10} \\
 i_{R9} + i_{R11} &= i_{R12} \\
 i_{R10} + i_{R12} &= i_t
 \end{aligned}$$

$$\begin{aligned}
 V_{R1} &= S_1 - S_2 = i_1 R_1 = S_1 - S_2 \\
 V_{R2} &= i_2 R_2 = S_2 - S_3 \\
 V_{R3} &= i_3 R_3 = S_1 - S_4 \\
 V_{R4} &= i_4 R_4 = S_2 - S_5 \\
 V_{R5} &= i_5 R_5 = S_3 - S_6 \\
 i_{R6} &= S_4 - S_5 \\
 i_{R7} &= S_5 - S_6 \\
 i_{R8} &= S_4 - S_7 \\
 i_{R9} &= S_5 - S_8 \\
 i_{R10} &= S_6 - S_7 \\
 i_{R11} &= S_7 - S_8 \\
 i_{R12} &= S_8 - S_9
 \end{aligned}$$

AX = y

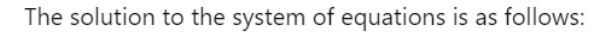
Touchpad – Network 2 in the demo board



1. $i_t = i_1 + i_3$
2. $i_1 = i_2 + i_4$
3. $i_3 = i_6 + i_8$
4. $i_2 = i_5$
5. $i_8 = i_{11}$
6. $i_4 + i_6 = i_7 + i_9$
7. $i_5 + i_7 = i_{10}$
8. $i_9 + i_{11} = i_{12}$
9. $i_{10} + i_{12} = i_t$
10. $i_1 \cdot r_1 = s_1 - s_2$
11. $i_2 \cdot r_2 = s_2 - s_3$
12. $i_3 \cdot r_3 = s_1 - s_4$
13. $i_4 \cdot r_4 = s_2 - s_5$
14. $i_5 \cdot r_5 = s_3 - s_6$
15. $i_6 \cdot r_6 = s_4 - s_5$
16. $i_7 \cdot r_7 = s_5 - s_6$
17. $i_8 \cdot r_8 = s_4 - s_7$
18. $i_9 \cdot r_9 = s_5 - s_8$
19. $i_{10} \cdot r_{10} = s_6 - s_9$
20. $i_{11} \cdot r_{11} = s_7 - s_8$
21. $i_{12} \cdot r_{12} = s_8 - s_9$

With the conditions:

- $r_1 = r_2 = r_3 = r_4 = r_5 = r_6 = r_7 = r_8 = r_9 = r_{10} = r_{11} = r_{12} = 100$
- $s_1 = 3.3$
- $s_9 = 0$



- The intermediate variables are:

- $s_2 = 2.2$
- $s_3 = 1.65$
- $s_4 = 2.2$
- $s_5 = 1.65$
- $s_6 = 1.1$
- $s_7 = 1.65$
- $s_8 = 1.1$